When Pollution Rises, Reporting Falls: Coal Mining and Self-Monitoring under the Safe Drinking Water Act

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Abstract

Many regulatory regimes require regulated firms to self-report their own conditions, placing the cost of monitoring on the firm and creating an incentive to shirk on reporting. That incentive grows as monitoring costs rise, so if costs rise precisely in states of regulatory concern, firms under-report exactly when reporting is most valuable to the regulator. I document this perverse incentive among drinking water utilities in the United States. Coal mining is a potential source of water contamination, and sulfur regulation under the Acid Rain Program (ARP) drove exogenous shifts in the geography of coal extraction, altering regional contamination from upstream mining and raising the costs of both testing and treatment. To measure utility exposure to mining, I adapt an instrument for coal production based on the reallocation of coal supply across high- and low-sulfur mines following the ARP. Utilities with more upstream mining show higher arsenic, selenium, and barium concentrations, yet no increase in recorded contaminant limit violations. Instead, self-reporting of water quality falls by seven to ten percentage points, even as reporting requirements tighten with contamination. Regulators substitute their own monitoring for part of utilities' self-reporting responsibilities: an additional upstream mine raises the likelihood of a regulator visit by 14 percentage points, yet those visits uncover no contaminant limit violations. Sanctions do not rise to deter under-reporting; an additional upstream mine reduces the likelihood of formal enforcement by 5.7 percentage points. Regulators could therefore raise self-reporting by increasing penalties for under-reporting.

Keywords: self-reporting; monitoring and enforcement; environmental regulation; drinking water quality; Safe Drinking Water Act; coal mining; acid mine drainage; Acid Rain Program.

JEL classification: D82; K42; L95; Q48; Q53; Q58.

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1 Introduction

Between 9 and 34 million Americans each year are served by a drinking water utility that violated a health-based drinking water standard (Allaire, Wu, and Lall 2018). That count is assembled from utilities' self-reported water quality, and its reliability depends on utility self-reporting behavior when contamination in their systems is rising. As with other environmental regulations that involve self-reported compliance, utilities have an incentive to under-report in order to reduce the cost of self-reporting or to conceal a contaminant violation. This paper shows that when contamination rises, under-reporting increases to reduce the cost of self-reporting. Coal mining upstream of a utility's water intakes raises the arsenic, selenium, and barium concentrations measured in its water, yet produces no detectable increase in recorded contaminant limit violations. Instead, it reduces the likelihood a utility self-reports water quality by seven to ten percentage points. Since utilities provide drinking water to 85 percent of US households, it is pertinent to public health to understand the causes of compliance with water quality self-reporting.

How do firms and regulators respond to increases in contamination under a contamination control regulation that is monitored through self-reported compliance? Compliance monitoring is the first step in enforcing regulation, so a first-order question is how self-monitoring responds when contamination increases. In this paper, I identify the response of firms and regulators to increasing contamination of a firm's regulated output in the context of US drinking water utilities exposed to an exogenous increase in water contamination from coal mining between 1985 and 2005.

Regulators, faced with constrained resources, favor self-reporting, which shifts the information-gathering costs onto the regulated. However, self-reporters lack the incentive to disclose when the penalties for revealed non-compliance exceed the penalties for non-disclosure, and when self-reporting itself requires costly effort. Regulators generate compliance through a combination of inspections and penalties. Successful inspections identify concealed violations, but any inspection shifts the cost of monitoring onto regulators, the efficiency of which is discussed in the literature on fiscal federalism. Successful penalties offset the firm's incentives to under-report and contaminate at levels that regulators deem unsafe. When the costs of self-reporting increase as contamination increases, for example, due to increasing self-reporting requirements, the firm's incentives to under-report also increase. Enforcement needs to keep up with the incentives to under-report. Contamination control regulations that increase self-monitoring requirements without increasing penalties for under-reporting can induce under-reporting, undermine the regulation's monitoring, and raise regulators' costs through inspections.

US drinking water quality has improved substantially since the passage of the Safe Drinking Water Act ("SDWA") (Keiser and Shapiro 2019a), yet notable exceptions remain, such as the crisis in Flint, Michigan (Christensen, Keiser, and Lade 2023), and there are still large health costs associated with contaminated drinking water in the US (Hill and Ma 2017; Keiser and Shapiro 2019b; DiSalvo and Hill 2024; Marcus 2025). Some of this literature relies on utility self-reported water quality and excludes under-reported water quality. A related body of literature holds that the remaining gains from abating point-source pollution in the modern era are small. Bingham et al. (2000) find that point-source pollution is already well regulated in the US and that achieving zero emissions from point sources would improve only 10 percent of US rivers and streams in EPA quality ladder rankings. The benefits of surface water regulation may nonetheless be undervalued because they are difficult to measure, a problem that instrumental variables can address (Keiser 2019).

The physical mechanism linking coal mining to water contamination is well established in the geochemical literature (Skousen, Ziemkiewicz, and McDonald 2019; Akcil and Koldas 2006). Economic work documents coal mining's effects on air quality, greenhouse gas emissions, and health (Muller, Mendelsohn, and Nordhaus 2011; Parfitt 2024). This work has not established whether mining contamination reaches treated water at drinking water utilities and, through them, households. The question is not only historical: US federal energy policy under President Donald Trump has expanded mining access on federal lands and intervened to prevent planned coal-fired power plant retirements (U.S. President (Trump) 2025a, 2025b, 2025c; Daly 2025).

Firms' motivations to under-report have largely been studied as concealment. In work on water utilities, under-reporting has been attributed to concealing a contamination violation (Bennear, Jessoe, and Olmstead 2009; Naylor and Deaton 2025; Andarge et al. 2025; Wallsten and Kosec 2008; Foster et al. 2019), and the same motivation underlies under-reporting at oil and gas firms (Lee 2020). A related literature finds that regulated entities exploit gaps in monitoring coverage to evade contamination limits. Zou (2021) shows that ambient air quality deteriorates on days when monitors are not scheduled to operate, and Mu, Rubin, and Zou (2026) document strategic shutdowns of pollution monitors. Small cement plants are exempt from air quality reporting, so their emissions go unmonitored even though they cause large health damages (Zirogiannis et al. 2023). A related literature asks how to induce firms to self-report and what constitutes efficient self-reporting (Innes 1999; Foster and Gutierrez 2013; Anton, Deltas, and Khanna 2004), and Malik (1993) finds that infrequent visits combined with high penalties induce better self-reporting. This literature has not asked whether the cost of self-reporting itself, which rises with contamination when monitoring requirements increase, drives under-reporting; that cost channel has been identified in tax regulation (Tazhitdinova 2018) but not in environmental compliance monitoring.

The literature on the economics of enforcement and monitoring of environmental regulation (Shimshack 2014) finds that enforcement is effective when its severity is non-linear and higher violations receive higher enforcement (Harford 1987). Attention has centered on dynamic enforcement that increases fines when firms are repeat violators. Firm compliance improves when regulators target repeat offenders and dispense stricter enforcement to them (Blundell, Gowrisankaran, and Langer 2020; Blundell 2020). Duflo et al. (2018) find that regulators use inspections selectively to target the worst polluters, and that targeting enforcement increases future abatement. These studies condition enforcement on an observed violation. When contamination rises but the violation is concealed rather than recorded, enforcing the regulation is no longer possible.

This paper makes three contributions. First, this is the first study to identify the cost of self-reporting, rather than the concealment of a contaminant exceedance, as a cause of under-reporting by drinking water utilities. Utilities face a mandated increase in testing frequency when nitrate levels cross half of the contaminant limit. Under-reporting jumps immediately after that threshold is crossed, where readings are still below the limit, and there is no exceedance to conceal. Second, this is the first study to identify the effect of contamination itself on firm self-reporting and regulator enforcement. Enforcement does not intensify at the utilities that reduce self-reporting as contamination rises; regulators substitute their own sanitary visits for the utility's self-reporting obligations, absorbing the monitoring cost that self-reporting was designed to shift onto the regulated. Despite regulators increasing the frequency of visits to under-reporting utilities, they do not discover concealed contaminant exceedances. This provides additional evidence that under-reporting is driven by self-reporting costs. Third, it provides the first causal estimates of coal mining's effect on drinking water quality at treated utilities in the United States, identifying that coal mining pollution contaminates piped household water. Methodologically, I adapt the instrument for coal production of Douglas and Wiggins (2015), who use the sulfur-driven reallocation of coal demand under the Acid Rain Program to predict production at coal mines, and apply it for the first time to instrument a drinking water utility's exposure to coal mining through their water intakes. The results are applicable beyond drinking water. Self-reported compliance underpins regulations as diverse as the income tax, and wherever the cost of reporting rises with the quantity being reported, monitoring weakens precisely when it matters most.

Empirically identifying the impact of contamination on firm behavior requires separating the firm's choices over output and contamination. Separating a firm's output and contamination in most industries is challenging since firms can reduce contamination by reducing output. For example, in the US, the Toxics Release Inventory requires firms to report emissions, handling, and use of chemicals above a reportable quantity. Firms can avoid the reporting requirement by emitting less or producing less (Zou 2021; Mu, Rubin, and Zou 2026). The contamination of drinking water at utilities from coal mining is the ideal setting to study the effect of contamination on self-reporting since the main source of water contamination that utilities face arises upstream as an externality of a production process that the utility does not directly control.

Two sources of endogeneity need to be addressed to identify the causal effect of upstream coal mining on utilities. First, coal mines are located in areas with particular geological characteristics, such as coal seam accessibility and transportation corridors, that may also independently predict water quality and utility characteristics. Second, vulnerable utilities may receive regulator support to protect their source water and block upstream mine siting. Utilities that successfully resist upstream mines may be worse at maintaining compliance, generating a negative correlation between upstream mining and utility ability that understates the true contamination from mining.[1]

I address these identification challenges by instrumenting for the number of coal mines upstream of a drinking water utility's water intakes, using the shift in the location of coal production caused by the Acid Rain Program (ARP). The ARP is federal legislation that forced coal-fired power plants, the main consumers of coal, to reduce sulfur dioxide emissions by 1995. Plants complied by switching from high-sulfur coal, with sulfur greater than two percent by weight, to low-sulfur coal, with sulfur less than two percent by weight. The ARP created a nationally determined shift in the demand for each mine's coal based on its sulfur content and independent of the characteristics of downstream drinking water utilities. I construct a panel of drinking water utilities downstream of active coal mines from 1985 to 2005. I measure the sulfur levels of coal and the annual coal production occurring in the watershed upstream of a utility's water intakes. Utility behavior is measured by drinking water quality violations and self-reporting violations.

I formalize the utility's reporting incentives in a model in which a utility draws a compliance cost and a self-reporting cost, chooses how far to neglect its contamination control obligations, and then chooses to self-report water quality if costs are below a reporting cost cutoff and under-report otherwise. Contamination increases the probability

1. For example, the EPA Source Water Protection program can designate areas where drinking water is sourced, especially for utilities with poor water quality performance, and take an inventory of pollution from human activities or natural sources that pose the greatest threat to source water, making it difficult to site pollution-producing activities in these areas.

of higher cost draws from both cost distributions. The model yields three results that organize the empirical work: negligence rises with compliance costs, the share of utilities that under-report rises with contamination, and raising every sanction proportionately increases self-reporting. The second result is the paper's central prediction, and it holds through two channels: an increase in self-reporting costs against a fixed reporting cutoff, and a shift of the population toward high-cost types whose cutoffs are already low.

I document that coal mining raises contaminant concentrations measured at a subset of utilities between 1998 and 2005. Comparing utilities that experienced more upstream coal mining to those that experienced less upstream coal mining, I find that every additional 10 million short tons of coal mined upstream since 1985 increases mean arsenic, selenium, and barium concentrations by 79 percent, 69 percent, and 23 percent of their average concentration levels in the sample.

Coal mining significantly raises the incidence of under-reporting but has no detectable effect on contaminant concentration violations. In particular, coal mining reduces utility self-reporting of water contamination from inorganic chemicals ("IOC"), arsenic, and nitrates by seven to ten percentage points. Utilities receive a self-reporting violation and not a water quality violation when they do not self-report water quality. Since self-reports are required to determine a contamination exceedance violation, the lack of violations does not, by itself, demonstrate that coal mining does not lead to violations.

I investigate whether utilities reduce self-reporting rates following mining contamination due to the rising costs of self-reporting or to conceal a contamination exceedance violation. I find that some reduction in self-reporting occurs due to the costs of self-reporting. Water utilities are required to increase self-monitoring for nitrates when nitrate concentrations exceed 50% of the contamination limit. I estimate that the likelihood a utility fails to self-report nitrate concentrations rises by 26 percentage points within six months, and by 59 percentage points within a year, of a reading above 50% of the contamination limit. These results are not causal, and the direction, rather than the magnitude, should be interpreted as suggestive evidence of under-reporting driven by the costs of self-reporting.

I investigate regulator enforcement against utilities that experience rising contamination by measuring regulator inspections and enforcement. As contamination increases and self-reporting falls, regulators substitute utility self-reporting with regulator visits, shifting the burden of monitoring the regulation onto the regulators. An additional active coal mine upstream of a utility increases the likelihood that a utility receives a sanitary visit from a regulator by 14 percentage points. Visits nonetheless reveal no contamination violations. This is evidence that utilities are not under-reporting to conceal contaminant exceedances.

Regulator enforcement does not increase as contamination and under-reporting increase. I find that an additional active coal mine reduces the likelihood that a utility receives formal enforcement by 5.7 percentage points, even as the likelihood of receiving an enforcement visit rises by 3.7 percentage points. Enforcement visits do not always translate into enforcement, such as fines and court-ordered installation of water treatment technology. Regulatory attention therefore rises without the sanctions that would offset utilities' incentives to under-report as contamination rises.

The rest of the paper proceeds as follows. Section 2 describes the SDWA, the self-reporting and enforcement institutions it creates, and the Acid Rain Program. Section 3 sets out the model of negligence and self-reporting and derives its three propositions. Section 4 describes the data. Section 5 estimates the effect of coal mining on measured contaminant concentrations. Section 6 constructs the instrument and assesses its validity. Sections 7 and 8 report the effects of coal mining on utility self-reporting and contaminant limit exceedances, and on regulator visits and enforcement. Section 9 discusses the mechanisms and Section 10 concludes.

2 Institutional background

2.1 Drinking water utilities, regulators, and the Safe Drinking Water Act

I focus on drinking water utilities downstream of coal mines between 1985 and 2005. In particular, I focus on a class of drinking water utilities regulated by the SDWA and known legally as community water systems. These systems serve at least 25 year-round residents or have at least 15 service connections. The SDWA is a federal regulation that requires all community water systems to remove contaminants and self-report compliance to state regulators. Regulators issue a maximum contaminant level ("MCL") violation to a utility that self-reports a contaminant above a contaminant-specific limit. MCL violations are serious threats to public health and require timely rectification, and they often trigger enforcement. Regulators cannot issue MCL violations until a utility self-reports, creating an opportunity for utilities to conceal an MCL violation by not reporting, which I refer to as under-reporting, and receive a monitoring and reporting ("MR") violation. Utilities have an additional incentive to under-report to avoid self-reporting costs, which are expensive in and of themselves and require hiring technical and administrative staff and paying laboratories. Utilities found self-reporting requirements difficult to meet due to the administrative and technical burden. Utilities needed to be administratively organized to conduct the right tests and report them on schedule. Technical staff were required to conduct separate tests for contaminants or to prepare samples to be sent to laboratories. A survey of Indiana drinking water utility managers in 1980 found that the reporting requirements of the SDWA had the greatest impact on their systems (Oleckno 1982).

The number of tests a utility was required to conduct varied according to the regulator-assessed risk of contamination to its water supply. Utilities assessed as having impaired water were required to self-report more tests than a utility with less impaired water. Self-reporting requirements, and therefore self-reporting costs, increase with contamination. Contamination increases the likelihood that a utility's water quality exceeds the MCL, increasing the incentive to under-report in order to conceal an MCL violation.

The states and tribes enforce the SDWA and are responsible for bringing violating utilities into SDWA compliance. With the exception of a subset of violating circumstances and mandatory notifications to consumers that utilities must provide when in violation, the SDWA gives states wide discretionary powers to tailor compliance and enforcement plans to each utility and violation. The intention is for the violation to be resolved quickly enough to reduce the health impacts of contamination. MCL violations are treated seriously and receive stricter enforcement than MR violations. However, an intentional MR violation concealing an MCL violation is punished more harshly than a regular MR violation. Enforcement is frequently remedial rather than punitive in addressing the cause of the violation. Enforcement can be informal (letters, notices of violation, compliance plans) and formal (court mandated consent decrees, legal prosecution, fines). Non-financial penalty enforcement is still costly to utilities. Public notifications for violations have large reputation costs that deter violations (Bennear and Olmstead 2008). Consent decrees can require installing expensive treatment infrastructure or hiring technical personnel. The replacement of an existing arsenic filtration system at a utility with baseline arsenic levels of 0.029 mg/L, serving 250 people in Pennsylvania in 2011, cost $191,970, including $136,744 for equipment, $21,726 for site engineering, and $33,500 for installation. In addition, it cost an average of $60 a day in chemical, electricity, and labor costs (U.S. Environmental Protection Agency 2011).

Regulators also enforce utility compliance through on-site visits, which the EPA categorizes into five groups based on the reason for the visit: sanitary, technical assistance, enforcement, sample collection, and inspection visits (U.S. Environmental Protection Agency 2026b). The most extensively documented of these is the sanitary survey, an on-site evaluation, conducted by the state or tribal agency, of eight components of a utility's system: its water source, treatment process, distribution system, finished-water storage, pumps and pumping controls, monitoring and reporting practices, system management and operation, and operator compliance with certification requirements (U.S. Environmental Protection Agency 2026a). Utilities receive a sanitary survey at least once every three years; the survey also gives the primacy agency an opportunity to educate operators on proper monitoring and sampling procedures and to provide technical assistance (U.S. Environmental Protection Agency 2026a). Technical assistance visits involve engineering plan review and operations-and-maintenance guidance, and sample collection visits involve the regulator collecting the water sample.

Enforcement visits accompany a formal enforcement action, an investigation of a complaint or violation, or an emergency response, while inspection visits comprise site inspections, regularly scheduled inspections, and informal inspections outside the sanitary-survey cycle (U.S. Environmental Protection Agency 2026b). A 2024 EPA fact sheet on utility inspections describes the general on-site inspection process: inspectors open with a conference, walk through the facility to examine the source, treatment, distribution, and storage components and the condition of equipment, collect water samples, and review the utility's monitoring plans, sampling logs, operational manuals, and distribution system maps (U.S. Environmental Protection Agency 2024). Inspections may also follow up on a prior inspection, sanitary survey, enforcement action, or citizen complaint; EPA typically issues an inspection report within 60 to 90 days, and further enforcement action may follow if violations are identified (U.S. Environmental Protection Agency 2024).

I focus on the SDWA regulation of arsenic, nitrates, barium, selenium, and general IOCs since they can be contaminant byproducts of coal mining and their MCLs did not change between 1985 and 2005.[2] Nitrate and arsenic self-reporting requirements changed between 1985 and 2005; however, they changed nationally. Year fixed effects in empirical models are used to address the concern that changes in MR violations would be the result of changing self-reporting requirements. The SDWA IOC contaminant rule contains all regulated IOCs during this period, some of which are subject to changing MCLs and some of which may be contaminant byproducts of coal mining.

Arsenic, nitrates, barium, and selenium are commonly produced during coal mining through acid mine drainage ("AMD") (Liang et al. 2017; Boente, Baragaño, and Gallego 2020; Rouhani, Skousen, and Tack 2023). AMD occurs when rocks containing sulfides are exposed to air and water, beginning the process of oxidation and decomposition (Ceto and Mahmud 2000). The oxidized minerals form sulfuric acid, which dissolves metal ions and sulfates that can then be carried into surface or ground water (Akcil and Koldas 2006; Skousen, Ziemkiewicz, and McDonald 2019). Mines can reduce AMD by covering

2. The list of regulated contaminants has been added to over time: in 1985, there were 23 regulated contaminants, and by 2005, there were 93.

excavated material with water in tailing ponds. Coal mining can also generate pollution through coal wash slurry. Coal washing processes coal with water and chemicals to remove impurities and improve combustibility. The slurry waste is often mixed with water and stored in retaining ponds or abandoned mine shafts. Dissolved pollutants enter groundwater and surface water by seeping through porous rocks or through retaining pond failures (Sealey 2000).

2.2 The Acid Rain Program (ARP) on utility exposure to coal mining

I use the ARP to obtain a causally valid estimate of the response of utilities and regulators to coal mining. The ARP was signed on November 15, 1990, as an amendment to the Clean Air Act, with the intention of reducing acid rain by limiting emissions of acid rain precursors from coal-fired power plants: sulfur dioxide (SO2) and nitrogen oxides (NOx). The regulation capped total national SO2 and NOx emissions. Power plants were given emission allowances, for which they were fined if they exceeded the limit. Although compliance could be achieved with trading allowances, SO2 scrubbers, and NOx scrubbers, fuel switching to low-sulfur coal was the primary abatement response. The ARP achieved 100 percent compliance (Lattanzio 2022).

Emissions were capped in two phases. In Phase I, the EPA issued emission permits and allowances to 110 highly polluting, mostly coal-fired electricity-generating facilities. Facilities targeted under Phase I were required to achieve compliance with SO2 emission reductions by 1995. Phase II included smaller power-generating units that used oil, coal, and gas. The deadline for compliance with Phase II was January 1, 2000. I focus on the Phase I deadline because it targeted the largest coal-fired power plants, which were the largest customers for coal since the electric power industry accounted for 90 percent of coal demand in 2000 (Freme 2000). The ARP increased the demand for low-sulfur coal and reduced the demand for high-sulfur coal (U.S. Environmental Protection Agency 2001; Douglas and Wiggins 2015). Coal mines with high sulfur have a limited ability to post-process the coal to remove sulfur since part of the sulfur is integral to the chemical structure of the coal and cannot be removed (Wheelock and Markuszewski 1984).

3 Theoretical model

3.1 Setup

I build a theoretical model to formalize a utility's decisions to comply with water contamination regulation under rising contamination. Utilities are required to meet contamination requirements and self-report compliance. I follow Mookherjee and Png (1994) and assume that utilities find contamination compliance costly. Compliance cost is determined by a utility's ability, such as the skills of its managerial and technical staff, and the costs of water treatment, including the cost of filtration equipment and sterilization chemicals. A utility's compliance cost, θ, is a random variable drawn from $F(\cdot\,; m)$, a strictly increasing and continuously differentiable cumulative distribution function with support $(0, \bar{\theta})$, density $f(\cdot\,; m)$, and contamination m. A larger θ represents higher compliance costs. Increasing m weakly lowers the probability that a utility's compliance cost falls at or below any given level, so that $\partial F(\theta; m)/\partial m \leq 0$. Utilities know their draw of θ, but regulators only know the distribution of θ.

Utilities receive a benefit from neglecting their contamination control requirements. The utility chooses a negligence level $a \in [0, \bar{a}]$, which measures how much it avoids the requirements of drinking water quality regulations, from avoiding every requirement $\bar{a}$ to full compliance 0. Regulators do not directly observe a utility's choice of a. A utility receives a benefit $\theta b(a)$ from having compliance costs θ and choosing a negligence level a, where $b'(a) > 0$ and $b''(a) < 0$. A utility's benefits from negligent contamination control are the savings on wages and chemicals. The opportunity cost to the utility of choosing a compliance level a relative to maximum negligence $\bar{a}$ is $\theta[b(\bar{a}) - b(a)]$. The financial benefit of a certain level of negligence to a profit-maximizing, privately owned utility is obvious. Publicly owned utilities also benefit from cost savings that allow them to forgo rate increases or tax collection, which may be unpopular with residents.

Utilities are legally required to self-report their water quality each year but have the choice to under-report. Given negligence a, a year of self-reporting would reveal a contaminant exceedance with probability $q(a)$, with $q(0) = 0$, $q' > 0$, and $q'' \geq 0$. I modify the penalty structure from Kaplow and Shavell (1994) by assuming that self-reporting is costly. A utility's annual self-reporting cost $c \sim G(\cdot\,; m)$ is drawn from a strictly increasing and continuously differentiable cumulative distribution function with density $g(\cdot\,; m)$ and support $[0, \bar{c}]$, where the support itself does not vary with m. Regulators require utilities to increase sampling as contamination increases, so the distribution $G(\cdot\,; m)$ depends on upstream contamination m with $\partial G(c; m)/\partial m \leq 0$ for every c. Raising m weakly lowers the probability that a utility's self-reporting cost is at or below any given level.

Regulators perfectly observe under-reporting. Unlike Mookherjee and Png (1994), I assume regulators do not observe contaminant exceedances unless utilities self-report water quality or the regulator conducts an inspection. Regulators use penalties and investigations for under-reporting and contaminant exceedances. Utilities receive a sanction r if they self-report a contamination level above the MCL and a penalty t for under-reporting. Regulators investigate under-reporting utilities with a probability p and issue a sanction s for a discovered contaminant exceedance, where $s > r$.

I maintain two assumptions on the sanctions throughout. First, $r > ps$: the certain sanction a utility bears for disclosing a contaminant exceedance exceeds the expected sanction it bears for concealing one, which is what gives a utility a reason to conceal in the first place. Together with $s > r$ this requires the investigation probability to satisfy $p < r/s$. Second, $t \geq (r - ps)\, q(\bar{a})$, which keeps the self-reporting threshold defined below non-negative at every negligence level. Without it, $(r - ps)q(a)$ could exceed t at high negligence, and since self-reporting costs satisfy $c \geq 0$, a negative threshold would mean no utility with any cost draw self-reports.

I maintain one regularity condition on the negligence choice: b and q are such that optimal negligence is interior, $a \in (0, \bar{a})$, at every type in the support of θ and at either marginal sanction, r or ps. Interiority makes the first-order condition below both necessary and sufficient within a decision to self-report or under-report.

Each year the utility chooses to self-report at a cost c or not report and receive an under-reporting violation. Expected costs are the sum of expected penalties and self-reporting costs

$$\begin{aligned} \text{self-report:} \quad & c + q(a)\, r \\ \text{under-report (MR):} \quad & t + p\,[q(a)\, s] \end{aligned}$$

The utility self-reports if and only if

$$c \leq \hat{c}(a) \equiv t - (r - ps)\, q(a). \tag{1}$$

$(r - ps)q(a)$ is the wedge between self-reporting a contaminant exceedance and concealing it. Even a system with no MCL violations, $q(a) = 0$, stops self-reporting once $c > t$, when self-reporting is more expensive than the expected sanction for skipping it, and there is no MCL violation to conceal. The self-reporting threshold is decreasing in $q(a)$, so systems with a higher probability of exceeding the MCL reduce self-reporting at lower cost thresholds. The expected annual cost of the reporting decision, at a cost draw c, is

$$e(a; c) = \min \{ c + q(a)r, \quad t + q(a)ps \} , \tag{2}$$

which is increasing in a.

3.2 Model predictions

The model predicts that the share of utilities failing to self-report rises with contamination, and it does so through two channels, neither of which requires a utility to be hiding a contaminant exceedance.

The first is a pure cost channel. A utility self-reports when its draw of the self-reporting cost falls below the cutoff $\hat{c}(a)$ in equation (1), which compares the cost of a round of testing against the penalty for skipping it, net of what the utility saves by not revealing an exceedance. Regulators require more frequent sampling of utilities whose source water is impaired, so contamination shifts the distribution of self-reporting costs to the right while leaving each utility's cutoff where it was. More of the population lands above its own cutoff, and under-reporting rises. This channel operates even for a utility whose water quality does not approach the contaminant limit, since such a utility still stops reporting once self-reporting costs more than the penalty for not reporting, and it has no contaminant exceedance to conceal.

The second channel works through the composition of compliance costs. Contamination also raises the cost of treating water to any given standard, shifting the distribution of the compliance-cost type θ to the right. Utilities with higher compliance costs choose more negligence, and a more negligent utility faces a higher probability that a self-report would reveal an exceedance, which lowers its own reporting cutoff. Contamination therefore moves the population toward types that are already closer to not self-reporting. The comparative statics of both channels are non-negative, so the share of utilities under-reporting is non-decreasing in contamination. This is Proposition 2, and it is one of the predictions tested in the empirical work.

The model predicts that the probability of violating a contaminant exceedance increases when contamination increases, conditional on a utility conducting a self-reported water quality test. By assumption, contamination raises compliance costs. Compliance costs raise negligence since the benefits of negligence are scaled by the compliance cost. Finally, the probability of violating a contaminant exceedance upon self-reporting increases with negligence. Contaminant exceedances will not be detected in the data if utilities do not self-report water quality.

Appendix A states and proves the three propositions: negligence is increasing in compliance costs (Proposition 1), the share of under-reporting utilities is non-decreasing in contamination (Proposition 2), and scaling every sanction by a common factor raises self-reporting (Proposition 3). Proposition 3 is the model's policy statement and is not tested here.

4 Data

I measure the behavior of utilities and regulators under a contamination self-reporting regulation when contamination increases in a panel dataset of drinking water utilities downstream of at least one active coal mine from 1985 to 2005. Utility behavior and performance are measured with annual self-reporting violations, contamination exceedance violations, and contaminant concentration measurements. Regulator action is measured through enforcement and visits to the utility. A utility's exposure to coal mining is measured by taking the sum of active coal mines and the total coal produced at each mine in the watershed directly upstream of the utility's water source. I measure the characteristics of coal within watersheds by the percentage of its weight composed of sulfur. I describe in the instrumental variable section how coal's sulfur content is used to create an exogenous measurement of a utility's exposure to coal production.

4.1 Utility characteristics, regulator, and violation data

The EPA's Safe Drinking Water Information System ("SDWIS") Enforcement and Compliance History Online ("ECHO") provides data on the universe of utilities active between 1985 and 2005. This time period lacks time-varying utility characteristic data, except for data on the number of facilities such as water intakes and treatment plants operated by each utility. I use the number of active facilities as the only time-varying utility characteristic.

SDWIS ECHO provides data for utility MR violations, MCL violations, and regulator enforcement and utility visits. SDWIS records violations accurately but not exhaustively. A survey of 1,800 utilities in 2000 found that 10 percent of MR violations and 50 percent of MCL violations were in SDWIS but that 95 percent of violations in SDWIS were accurately reported (U.S. Environmental Protection Agency 2000).

I construct outcome variables for MR and MCL violations for a general IOC category, arsenic, and nitrates. SDWIS ECHO uniquely identifies utility violations with information on the start and end dates of the violation, the SDWA rule violated, and the contaminant. To avoid double counting violations, I exclude any IOC violations that occur at the same time and at the same utility as an arsenic or nitrate violation. I measure utility violations during a year with a binary variable equal to one if the utility experienced a violation of that type at any time during the year. I measure the number of days per year a utility was in violation as a robustness check because SDWIS ECHO often records violations as starting when utilities fail to meet regular testing schedules and then continue for a fixed period after they are issued rather than recording the precise dates a contaminant was elevated.

Table 1 presents summary statistics for two measurements of annual utility violations: whether there is any violation in a year and the number of days violated in a year. The summary statistics are calculated from utility-year cells. MCL violations rarely occur; a total of 26 MCL violations occur over the entire panel. MR violations occur several orders of magnitude more frequently than MCL violations; depending on the contaminant, between 250 and 377 MR violations occur during the panel. The mean length of MR violations of eleven to sixteen days out of the year conceals the bimodal distribution of the length of MR violations. When an MR violation occurs, it tends to last the entire year; the 90th percentile of the days in a year experiencing an MR violation is 0, and the 99th percentile is 365. The rarity of MCL violations results in a zero 99th percentile.

I measure regulator enforcement with a binary variable equal to one for every year a utility experienced enforcement. SDWIS ECHO enforcement records include the date enforcement began and ended, and whether it was formal or informal. The "Whole panel" columns of Panel A of Table 2 summarize the share and number of cells in the sample that a utility received an enforcement action. Any enforcement occurs 21% of the time; most of that enforcement is informal, which occurs 19% of the time compared to 3.5% of the time when formal enforcement occurs. A utility can experience both types of enforcement simultaneously so any enforcement is not the sum of informal and formal enforcement. Enforcement is more likely to occur during a violation; conditional on an active violation, the rate of enforcement is 2 to 3.5 times higher than the unconditional rate. The "During MR year" and "During MCL year" columns of Panel A restrict the sample to utility-year observations where an MR or MCL violation of any type occurs. The likelihood of any enforcement increases from 21% in the unrestricted sample to 71% and 65% for the samples restricted to utility years with an MR and MCL violation, respectively.

I measure regulator visits to utilities with a binary variable, which is equal to one during a year when a utility received a regulator visit and zero otherwise. SDWIS ECHO records the date of the visit and reasons for visits, including sanitary survey, technical assistance, enforcement, sample collection, and inspections. The "Whole panel" columns of Panel B of Table 2 summarize the share and number of cells in the utility-year sample where a utility received a visit from the regulator. Sanitary survey visits are the most common type of regulatory visit at 14% compared to the second most common, inspection visits at 4%. Technical assistance, enforcement, and inspection visits are generally more likely to occur during years when utilities receive an MR violation, increasing the likelihood by 1.4 to 2.1 times the unconditional rate. Sample collection visits are less likely to occur during years with an MR violation compared to other years. This is counterintuitive since MR violations occur precisely when utilities do not self-report their water quality, and sampling information would reveal contamination levels. Sanitary survey visits occur with a similar rate conditional on an MR violation and at all times. This suggests that regulators conduct more activities than just sampling water quality in response to utility under-reporting. Regulators rarely use visits to respond to MCL violations. Visits rarely coincide with MCL violations; two sanitary visits were the most that occurred during an MCL violation. I cannot draw a definitive conclusion given the infrequency of MCL violations.

4.2 Utility contaminant measurement data

I measure utility contaminant concentrations of arsenic, nitrates, barium, and selenium with data from the second Six-Year Review ("SYR2", sample years 1998–2005). Although this data only covers a subset of utilities and years, it allows me to test whether coal mining raises water contamination at downstream utilities.

Beginning in 1991, the EPA began conducting reviews of all National Primary Drinking Water Regulations every six years to determine whether the rules needed revisions. The SYR2 was the second round of the six-year review covering 1998 to 2005 and is the earliest six-year review round to be made publicly available. The SYR2 data contained contaminant data for 89 percent of the total population served by a drinking water system but excluded Pennsylvania, Mississippi, Kansas, and Louisiana because those states failed to submit records.

From the raw SYR2, I retain community water system utilities and drop transient and non-transient non-community water systems. This removes roughly 23 percent of raw monitoring records.

The data contains contaminant level observations, whether the contaminant was detectable, and the value of the detection. Non-detections vary based on lab equipment sensitivity. To avoid a non-detect being interpreted as a zero contaminant concentration, I clean the SYR2 data following the procedure of Ravalli et al. (2022). For each contaminant, I compute the national share of records above the method detection limit ("MDL"), the minimum concentration at which a laboratory can reliably detect the contaminant in a sample, and I retain only contaminants detected in more than 10 percent of records. I replace the observations below the MDL of any contaminant with the MDL/$\sqrt{2}$; this prevents records from being dropped or mistakenly reported as zero without meaningfully inflating contaminant levels. Records reporting a concentration greater than 100 times the contaminant's MCL are dropped as data-entry errors. Among the IOCs that can be produced during coal mining, this keeps arsenic, nitrate, barium, and selenium. All mass-concentration readings are converted to mg/L.

After cleaning, sample-level records are collapsed to a panel at the utility $\times$ contaminant $\times$ year level. The annual concentration of a particular contaminant is the mean of that year's samples at a particular utility.

Panel A of Table 3 summarizes the contaminants included in my panel. The maximum value of all contaminant readings is below their MCL. The maximum arsenic reading of 0.0110 mg/L would exceed the stricter arsenic MCL of 0.010 mg/L, which took effect in 2006, but would not exceed the pre-2006 MCL of 0.05 mg/L. The fact that mean concentrations do not exceed the MCL is consistent with the low rates of MCL violations at utilities. Conditional on testing, the water contaminants appear to be within the regulatory limits. These summary statistics do not provide information on the contaminant concentrations that would be measured when a test was required by regulation but was not conducted by the utility. This data does not confirm that water contamination never exceeded the MCL. Most MR violations were issued for failure to test water quality. The SYR2 necessarily reflects the results from water quality tests and does not account for tests not undertaken that were subsequently punished with an MR violation.

SYR2 does not contain all test results submitted to regulators, but it is more likely to contain the test results of utilities that, at some point during my sample, fail to self-report their water quality and receive an MR violation. These results are more likely to reveal the contamination at the most negligent utilities, since the SYR2 represents an oversampling of more negligent utilities that are more likely to have below average water quality. Utilities that appear in the SYR2 data are more likely to violate the SDWA than those utilities that never appear in the data. Table 4 reports a two-sample t-test to compare the shares of utilities that appear in the SYR2 and have any SDWA violation to the share of utilities that never appear in the SYR2 data and have any SDWA violation. Utilities that appear in the SYR2 data are 18 to 23 percentage points more likely to experience any type of MR violation than utilities that never appear in the SYR2 data. The lack of SYR2 observations does not mean the utility was not self-reporting, since utilities can be compliant with MR rules and not have any SYR2 observations. The utilities that are more likely to self-report data are also less likely to have contaminant concentration observations in SYR2. Utilities in the SYR2 are between 1.3 and 3.7 percentage points more likely to experience any type of MCL violation compared to utilities that never appear in the SYR2 data.

4.3 River and stream connectivity data

Drinking water utilities are exposed to water pollution from coal mining when their source water is downstream of coal mines. The exposure is strongest when a utility draws from surface water. Groundwater sources downstream of coal mining are also exposed to upstream pollution since groundwater is recharged by water flowing from above. Groundwater sources are less exposed to contamination than surface water since the contaminated water flows through rock and substrate before recharging groundwater. I identify which utilities in the SDWIS ECHO national dataset source water from a watershed that is downstream of coal mining using the NHDplus hydrological connectivity dataset. The NHDplus hydrological dataset identifies the boundaries and connectivity of watersheds across the US. I combine it with an EPA dataset on the watershed-level location of utility water intakes.

The NHDplus assigns watersheds a unique 2- to 12-digit code known as a hydrologic unit code (HUC). The number of digits corresponds to the size of the area covered by the watershed, with the geographical area decreasing with the number of digits; for example, the lower 48 US states are divided into 18 HUC02s and over 87,000 HUC12s (see Figure 1). I use HUC12 watersheds to identify utility water intake locations. HUC12 watersheds are the smallest mapped watersheds available for the entire continental US and are small enough that coal mining activities may impact the water in the HUC12 directly downstream. To ensure that a utility intake is downstream of a coal mine, I exclude those utilities with an intake located in the same HUC12 as a coal mine; a utility water intake at a HUC12's inflow may not be affected by mining at the same HUC12's outflow. I limit my sample to utilities with at least one water intake in a HUC12 directly downstream from a HUC12 with a coal mine between 1985 and 2005.

4.4 Coal mine and coal sulfur data

My IV relies on predicting coal production from the sulfur content of coal and whether the time period is before or after the 1995 deadline for coal fired power plants to meet sulfur emission limits under the ARP. I measure the number of coal mines and the tons of coal extracted in the HUC12 watershed using individual coal mine production data from the Energy Information Administration ("EIA"). It includes the tons of coal produced by each mine in each year from 1985 to 2005. I merge EIA coal mine data with Mine Safety and Health Administration ("MSHA") data containing mine latitude and longitude coordinates. Where a utility has multiple intakes downstream of multiple HUC12s, I measure upstream coal production as the sum of production across those HUC12s.

The demand for coal from each mine depends on the physical characteristics of its coal. Thus, the coal production upstream of a utility depends on the average sulfur percentage in coal deposits in the watersheds upstream of the utility water intakes. Where a utility has water intakes in multiple HUC12s, I measure upstream sulfur as the mean sulfur level across its upstream HUC12s. I use the USGS COALQUAL dataset to measure the sulfur percentages present in coal within a HUC12. The USGS COALQUAL dataset consists of geospatially explicit drilled coal core samples from across the US. Samples measure characteristics, including the percentage of coal weight that consists of sulfur. I match COALQUAL samples to HUC12 boundaries that lie within 20 km of their coordinates, irrespective of whether coal production took place in that HUC12. This approximates coal deposits with the potential to be extracted. The mismeasurement of sulfur levels due to extreme variation within HUC12s, where some high-sulfur coal HUC12s may actually contain low-sulfur coal, is not an issue since HUC12s have low variation in measured coal sulfur levels. I drop HUC12s from my sample that are further than 20 km from a COALQUAL sample.

High-sulfur (greater than two percent sulfur) coal production fell between 1985 and 2005, while low-sulfur (less than two percent sulfur) coal production was more resilient and in some places increased. Figure 4 plots a histogram of the sulfur percentage of the coal bed within the HUC12s where a utility water intake and coal production are located. The sulfur content of HUC12 coal beds ranges from zero to 4.5 percent, with most of the HUC12 watersheds below 1.5 percent. Slightly more than half of the HUC12s are low sulfur coal beds.

In Figure 2, the HUC12s that contain coal production and are upstream of a utility water intake are green, while the HUC12s that contain coal production and are not upstream of a utility are red. The highest number of watersheds in my sample is located in Pennsylvania, Kentucky, and West Virginia. The eastern coal fields more frequently occur in HUC12s upstream of drinking water utilities. Western coal production takes place more frequently in watersheds without utility water intakes and is less well represented in my sample.

Figure 3 plots coal production from high-sulfur and low-sulfur HUC12 watersheds from 1983 until 2005. The mean quantity of coal produced in HUC12s with low-sulfur coal increased over this period, while it fell in HUC12s with high-sulfur coal. The same trend appears in the total national quantity of coal produced in HUC12s with low-sulfur coal. The average number of active coal mines within HUC12s with low-sulfur coal fell until 1995, when it stabilized and slightly increased. The number of coal mines in HUC12s with high-sulfur coal fell continuously over the same period.

Coal characteristics vary within coal regions, so the relative differences in coal production between high- and low-sulfur coal resulted in within-region spatial variation in which HUC12 watersheds experienced increases and decreases in production. Figure 5 plots the change in coal production at the HUC12 watershed level between 1985 and 2005, with circles sized proportionately to the change. Within states and regions, coal production rarely experienced a uniform decline or increase. Coal production mostly decreased in Pennsylvania and mostly increased in West Virginia. Other states' coal production saw both increases and decreases. Production increased and decreased in the western and eastern regions.

I use descriptive and summary data to show that the reduction in high-sulfur coal production also reduced the size of the population reliant on a utility exposed to coal production. These results do not reflect the actual population served by utilities affected by coal mines from 1985 to 2005 because I rely on the population served by a utility using the SDWIS snapshot from the third quarter of 2024. However, I have ongoing work to estimate the size of the population served by each utility from as early as 1990 using backcasting and machine learning.

The increases in coal production largely did not occur upstream of utility water intakes. Figure 6 plots the location and size of the population that obtains their drinking water from a utility directly downstream of a coal mining watershed. Circles are proportionate to the total population reliant on drinking water directly downstream of a coal mining watershed. Red circles indicate a population exposed to more upstream coal mining in 2005 compared to 1985; blue circles indicate a population exposed to less upstream coal mining in 2005 compared to 1985. The number of people exposed to coal mining through utility drinking water fell from 1.3 million in 1985 to 0.2 million in 2005. Coal mining increased in areas that largely did not provide source water to utilities.

5 The impact of coal mining on downstream utility contamination

5.1 Empirical framework

I measure the effect of coal mining on IOC water contaminant concentrations at utilities. This establishes whether coal mining is a convincing source of contamination to study the effect of contamination on utility and regulator behavior under a self-reporting contamination control regulation. I limit the estimation to arsenic, nitrates, barium, and selenium, as the MCL for these IOC contaminants did not change between 1985 and 2005, and these were the IOC contaminants that were retained after cleaning the SYR2 data.

Higher levels of cumulative coal production, the total amount of coal extracted since 1985, are associated with higher average contamination levels. Panel B of Table 3 measures the mean contaminant concentration readings at utilities that experienced a higher than median amount of cumulative tons of coal production, which is 94,000 cumulative short tons. Compared to the mean contaminant concentrations in the full sample of utilities in Panel A, utilities downstream of higher than median cumulative short tons of coal production have slightly higher concentrations of nitrates, arsenic, and barium.

In order to control for utility and regional effects that influence contaminant concentration levels, I estimate the impact of coal mining on the mean contaminant concentrations at utilities using the regression equation

$$Concen_{cht} = \beta \cdot \sum_{\tau=1985}^{t} \text{upstream coal prod}_{c\tau} + \rho_c + HUC02_h \cdot \tau_t + \epsilon_{cht} \tag{3}$$

$Concen_{cht}$ is the mean concentration of a contaminant measured at utility c in HUC02 h in year t using SYR2 data from 1998 to 2005. $\sum_{\tau=1985}^{t}$ upstream coal $\text{prod}_{c\tau}$ is the cumulative tons of coal mined in all HUC12s directly upstream of all utility c intakes from 1985 until year t. β is the effect of an additional 10 million cumulative short tons of coal mined upstream of utility c in year t since 1985. Utility fixed effects are included as ρ_c. To control for annual industrial and watershed level contamination trends that affected the entire watershed, I include HUC02 fixed effects interacted with time fixed effects $HUC02_h \cdot \tau_t$. Standard errors are clustered at the utility level.

Equation 3 identifies β from relative differences in exposure to cumulative coal production rather than comparing treated units to a control, since there are no utilities in my sample that are not downstream of coal production and can act as control observations. The identifying assumption requires that the contamination concentrations at utilities that experience a large decline in coal production would have trended the same as the pollution concentrations at utilities with a small decline.

I do not treat these estimates as causal. There is no instrument for cumulative upstream production here and no untreated comparison group. The concentration data cover a subset of utilities from 1998 to 2005 rather than the full panel, and the treatment is measured as cumulative tons of coal rather than as the count of active mines used in the rest of the paper. The section establishes descriptive evidence for how coal production affects utility contaminant concentrations. These results motivate using coal mining as a contamination shock to utilities.

5.2 Comparability of utilities by upstream production

Since the estimates of equation 3 are observational, it matters whether the utilities that received more upstream coal production already differed from those that received less. Table 10 regresses upstream production from 1998 to 2005 on utility characteristics. I do not have utility characteristic data that varies over time and so cannot include utility fixed effects. However, I include HUC02 fixed effects. Columns one and two use cumulative production, the intensive margin from which equation 3 is identified; columns three and four use an indicator for any upstream production.

On the intensive margin, utility characteristics are jointly insignificant, both with and without watershed fixed effects, with p-values of 0.31 and 0.75. Among utilities that have upstream mining, neither utility size, nor the number of water intakes a utility operates, nor whether it draws on surface water, nor whether it is publicly owned predicts how much coal is mined upstream. On the extensive margin, the characteristics are jointly significant, driven by the number of intake facilities: utilities operating more intakes are more likely to have any upstream mining at all, which is unsurprising since more intakes mean more watersheds in which a mine can sit. This motivates using the cumulative production as the measure of coal production in equation 3.

5.3 Results

Table 11 contains the results of equation 3 estimated on IOC contaminant concentration tests throughout utility water processing stages. The concentration sample is small, consisting of between 319 and 432 utility-year observations depending on the contaminant, because it is restricted to the utilities and years for which the Six-Year Review reports a reading. An additional 10 million cumulative short tons of coal mined upstream since 1985 is associated with higher mean concentrations of IOC contaminants at utilities. The association is significant for all IOC contaminants except for nitrates, which I lack the power to identify at a statistically significant level.

Arsenic increases by 0.0023 mg/L with each additional 10 million short tons of coal mined upstream. This is economically significant, equivalent to a 79 percent increase over the mean arsenic concentration of 0.0029 mg/L, and is statistically significant at the 1% level. The effect represents 23 percent of the arsenic MCL of 0.010 mg/L that took effect in 2006, and 4.6 percent of the 0.050 mg/L MCL in force before then. An additional 10 million cumulative short tons of coal increase selenium and barium levels by 0.0033 mg/L and 0.0171 mg/L, respectively. The effects are statistically significant at the 5% level for selenium and the 10% level for barium and are large relative to the mean contaminant levels, representing increases of 69 percent and 23 percent, respectively.

The results are more modest when scaled for the average amount of cumulative upstream coal production of 4.4 million short tons. With average cumulative upstream coal production, contaminant levels of arsenic, barium, and selenium increase by 35 percent, 10 percent, and 30 percent of their respective mean contaminant concentration levels.

An additional 10 million short tons of coal increases nitrate concentrations by 0.057 mg/L; however, the effect is not statistically significant. The effect represents a 7.6 percent increase in concentration levels relative to the mean contaminant concentration of 0.75 mg/L. This small increase could explain why the coefficient lacks significance. Assuming the contaminant concentration increases linearly with each additional 10 million short tons of coal mined, reaching 50 percent of the nitrate MCL from the sample mean would require approximately 743 million cumulative short tons of upstream coal production; the maximum cumulative amount of coal mined upstream of a utility anywhere in the sample is far below this.

Coal is associated with increases in contaminant concentrations at utilities downstream. The effects are economically and statistically significant, but within the sample, they are not associated with increases in contaminant concentrations above the MCLs. This is evidence that upstream coal mining can be used to measure the response of utilities to increases in water contamination.

6 An instrumental variable for upstream coal mining

I have demonstrated that coal production raises water contamination levels at drinking water utilities. I will use an IV for coal production to investigate the effect of contamination on utility self-reporting, contaminant limit exceedances, and regulatory enforcement. An IV is necessary to address the ability of utility characteristics to affect the spatial and temporal distribution of coal mines and drinking water utilities which, if estimated with OLS, would lead to biased coefficient estimates due to omitted variable bias. Utility characteristics that determine performance are likely correlated with the number of mines a utility is exposed to upstream. For example, a utility with inferior water treatment infrastructure that has limited ability to improve raw water may receive extra regulator support to protect its source water from coal mining. A utility with inferior water treatment infrastructure is more likely to violate water quality regulations than one with advanced water treatment infrastructure. This results in the number of upstream coal mines being negatively correlated with violations. A utility's ability is likely to change over the 20 year time span I study and cannot be controlled with utility fixed effects. Without data on utility ability, an OLS regression of utility behavior on coal production leads to an underestimate of the true effect of coal mining on utility behavior.

I adapt the IV from Douglas and Wiggins (2015) to predict the number of mines operating upstream of the utility based on the sulfur content of coal in the upstream watershed and whether the time period is before or after the 1995 deadline for coal-fired power plants to reduce sulfur emissions under the ARP.

I use a two-stage least squares regression, but the conceptual first stage is a regression of the number of active coal mines upstream of a utility on the sum of sulfur as a percentage of coal weight in each upstream watershed, interacted with a binary variable equal to one for all years after 1995 and zero otherwise. The first stage is

$$\begin{aligned}\text{NumMines}_{ct} = {} & \alpha_0 + \alpha_1 \text{NumFacility}_{ct} + \alpha_2 \text{UpstreamCoalSulfur\%}_c \cdot \text{Post95}_t \\ & + Y_t + C_c + \nu_{ct}\end{aligned} \tag{4}$$

The total number of mines directly upstream of utility c's intake in year t is $NumMines_{ct}$. The exogenous covariates are the number of utility facilities active in year t, $NumFacility_{ct}$, year fixed effects Y_t, and utility fixed effects C_c. AMD is more common when the minerals contain high levels of sulfur, making the incidence of AMD increase with coal sulfur levels (Schweinfurth 2009). Upstream sulfur is controlled for with utility fixed effects since sulfur levels do not change over time. The excluded instrument is the interaction between $UpstreamCoalSulfur\%_c$, the sum of sulfur as a percentage of coal weight in each watershed upstream of utility c, and a dummy variable equal to one for the years after the deadline to implement stage I of the ARP, 1995, and zero otherwise $Post95_t$. Standard errors are clustered at the utility level.

The coal sulfur content before and after 1995 predicts the amount of coal production well. Figure 7 plots the mean and standard deviation of HUC12 coal sulfur percentages by bins of the number of active coal mines in a HUC12 before 1995 and after 1995. In addition, it plots linear fit lines separately before 1995 and after 1995. There is a positive relationship between coal sulfur percentage and the number of coal mines before and including 1995. However, after 1995, the relationship flattened, and coal production heavily shifted towards coal with sulfur percentages below 1.5%. The change in production away from high-sulfur coal after the ARP is strongly demonstrated by the line of linear fit flattening after 1995, indicating that while there was more coal production in higher-sulfur coal areas before 1995, there was a large shift to areas with low-sulfur coal after 1995. The time-contingent switch in coal production based on sulfur content supports the structure of the variable $UpstreamCoalSulfur\%_c \cdot Post95_t$.

The strength of the instrument is still evident after controlling for covariates and fixed effects. Figure 8 is a scatter plot of the residualized number of upstream coal mines and the residualized instrumental variable. There is a negative relationship between the residualized number of upstream coal mines and $UpstreamCoalSulfur\%_c \cdot Post95_t$.

The first-stage is sufficiently strong. Table 5 reports the first-stage F-statistic, computed as the squared clustered t-statistic on the excluded instrument, which is numerically equivalent to the effective F-statistic of Montiel Olea and Pflueger (2013) in this just-identified setting. The F-statistic of 27.52 exceeds their critical value of 23.1 for a 5%-level test against 10% worst-case bias, well above the conventional rule-of-thumb threshold of 10 proposed by Staiger and Stock (1997).

The instrument requires monotonicity in the first stage to recover the local average treatment effect (LATE) estimate of the instrument.[3] In this setting, the instrument is $Z_c = UpstreamCoalSulfur\%_c \cdot Post95_t$, which equals zero before 1995 and $UpstreamCoalSulfur\%_c$ after. For a given sulfur level w, monotonicity requires either $\text{NumMines}_c(0) \leq \text{NumMines}_c(w)$ for all c, or $\text{NumMines}_c(0) \geq \text{NumMines}_c(w)$ for all c. This permits the direction of the response to vary with sulfur content. For example, at $UpstreamCoalSulfur\%_c = 1$ (low sulfur), the number of mines increases after 1995, so monotonicity requires $\text{NumMines}_c(0) \leq \text{NumMines}_c(1)$ for all c. At $UpstreamCoalSulfur\%_c = 4$ (high sulfur), the number of mines falls after 1995, so monotonicity requires $\text{NumMines}_c(0) \geq \text{NumMines}_c(4)$ for all c. These two conditions are compatible: monotonicity is satisfied so long as each pairwise comparison is uniform across units, even when low- and high-sulfur areas respond in opposite directions.

An implication of monotonicity is that there are no defier utilities. A defier utility

3. Stating monotonicity more formally and following Angrist and Imbens (1994), let Ψ be the support of the instrument Z_c, and for each $z \in \Psi$ define a random variable $D_c(z)$. The monotonicity assumption requires that for all $z, w \in \Psi$, either $D_c(z) \leq D_c(w)$ for all c, or $D_c(z) \geq D_c(w)$ for all c. This condition applies pairwise to elements of Ψ and imposes no restriction on the ordering of z and w.

experiences an increase in the number of coal mines it is downstream of after 1995 due to being downstream of a high-sulfur coal watershed, but would experience a decrease in the number of coal mines it is downstream of after 1995 due to being downstream of a low-sulfur coal watershed. This is unlikely since high-sulfur coal was more expensive than low-sulfur coal after 1995. To be a defier requires a power plant customer of a high-sulfur coal watershed to increase its coal demand after installing sulfur scrubbers and to increase its electricity market share. Power plants that rely on high-sulfur coal and scrubbers are likely to demand less coal since scrubbers are expensive to install, increase operating costs, require higher electricity rates to retain profits, and lose electricity market share. If some high-sulfur regions had a comparative advantage in low-cost production that offset the ARP cost increase, such as very low labor costs or transportation advantages, they might expand production even as ARP made their coal relatively more expensive. Such an expansion creates defiers only if it would not have occurred had those same low-cost regions instead contained low-sulfur coal; otherwise, it creates never-takers. A never-taker is a utility that is downstream of a low-sulfur watershed that experiences no increase in coal mines after 1995, and a utility that is downstream of a high-sulfur watershed that experiences no decrease in coal mines after 1995. A never-taker is not a violation of monotonicity and only reduces first-stage power.

The second stage of the IV regression is

$$\begin{aligned}\text{UtilityOutcome}_{ct} = \beta_0 + \beta_1\text{NumFacility}_{ct} + \gamma\widehat{\text{NumMines}}_{ct} \\ + Y_t + C_c + \epsilon_{ct}\end{aligned} \tag{5}$$

In the second stage, the dependent variable is a binary variable equal to one if utility c received an MR violation, MCL violation, regulator visit, or enforcement from the regulator in year t, $\text{UtilityOutcome}_{ct}$. Any national regulatory changes to the SDWA that affected all utilities are controlled for with time fixed effects. Since the sulfur content of a coal deposit is a fixed geological endowment, I control for it with utility fixed effects.

In addition to instrumental relevance, the IV produces causal estimates of the impact of coal mining on utility and regulator behavior if the instrument satisfies the independence and exclusion restrictions (Angrist and Pischke 2009). Conditional on utility and year fixed effects, the upstream coal production to which utilities are exposed through the instrument is independent of their potential outcomes because the scope of the ARP and the timing of its first phase were both determined independently of utility drinking water quality and utility characteristics. The instrument identifies variation in the location of coal production that is independent of unobserved utility characteristics and is based on the shift in coal-fired power plants' demand for lower-sulfur coal after 1995 in order to meet the sulfur dioxide emission standards of the ARP. The ARP was a national clean air policy that was not written or enacted to address water quality issues. The instrument exploits the national change in power plants' demand for coal by sulfur content, the timing and stringency of which were set without reference to any drinking water utility. The effects of the ARP on coal production were determined by national and international energy markets that local utilities or local utility water quality did not affect. Local utility characteristics regarding water quality or local power over source protection did not affect the scope of the ARP, the stringency of the ARP law, the degree to which low-sulfur coal should be part of the abatement technology, or the date of Phase I of the ARP. The timing of the ARP did not coincide with major drinking water regulations.

The instrument would violate independence if the unobserved utility characteristics that affect drinking water quality violations and the siting of coal mines changed after 1995 for utilities with different amounts of sulfur as a percentage of coal weight. Knowing about sulfur as a percentage of coal weight upstream of a utility after 1995 tells us nothing about how much political power a local utility has over mine siting.

Exogeneity requires the instrument to only affect utilities through its effect on the number of coal mines. A threat to this would be if utility SDWA regulation changed after 1995 and had a differential impact on utilities downstream of high-sulfur coal before and after 1995 compared to utilities downstream of low-sulfur coal before and after 1995. Changes in SDWA rules that affected all utilities in the same way are controlled for with time fixed effects. For example, in 1991, the monitoring and reporting schedule changed for all utilities and is controlled for with time fixed effects. Any change to an MCL would apply to all utilities and be controlled for with time fixed effects.

The exclusion restriction fails when the instrument affects regulator behavior through another path outside of its effect on coal mining. The exclusion restriction would be threatened if the ARP Phase I directly changed state EPA enforcement practices in a way that affected utilities differently according to the percentage of sulfur by weight in upstream coal. However, states were not primarily responsible for enforcing the ARP; it was a federally regulated program that, by itself, would not have affected state primacy enforcement of the SDWA.

Another exogeneity threat is local economic activity. The shift in demand away from high sulfur coal reduced mining employment and income in the communities that a utility serves, which could affect a utility's staffing and revenue, as well as a state regulator's enforcement priorities, without operating through water contamination. Absorbing this channel requires state-by-year fixed effects, which I plan to incorporate in the near future.

The exclusion restriction cannot be tested directly in a just-identified model, but individual alternative pathways can be tested. I test one such pathway by regressing the number of a utility's active intake facilities, a utility characteristic, on the instrument. This variable also enters both stages as an exogenous covariate, so evidence that the instrument moved it would indicate both an alternative pathway and that it is an inappropriate control. Table 6 reports the results. The coefficient on the instrument is small and statistically indistinguishable from zero, both in the specification with utility and year fixed effects and in a levels specification restricted to utilities observed throughout the sample period. This rules out one pathway; it does not establish the exclusion restriction.

Relevance, independence, and exclusion are sufficient for the IV estimates to produce causal estimates of the effect of coal mining on drinking water utility and regulator behavior. Relevance is verified above. Independence and exclusion are not testable; I argue for each on the institutional grounds set out in this section. Monotonicity is required not for causality but for interpretation. Under monotonicity the estimate is a local average treatment effect: the effect of an additional upstream coal mine at the utilities whose upstream mining responded to the ARP. Were monotonicity to fail, the estimate would still be assembled from causal effects, but it would weight defier utilities negatively and would no longer be an average effect for an identifiable group of utilities.

7 The impact of coal mining on utility self-reporting and contaminant limit exceedance

I now measure the effect of coal mining on utility self-reporting and contamination exceedance violations. The theoretical model predicts that coal mining causes utility self-reporting to fall and the probability of a contaminant exceedance to increase. Table 7 reports the results of equation 5, estimating the effect of an additional active coal mine on the likelihood that a utility receives either an MCL or MR violation for nitrates, arsenic, and inorganic chemicals using OLS, two-stage least squares, and reduced-form regressions.

Coal mining increases the likelihood a utility experiences an MR or MCL violation. Estimated with OLS, the effect of an additional active coal mine upstream increases the likelihood of an MCL or MR violation for nitrates, arsenic, and IOCs by 1.7 to 1.9 percentage points. The reduced form effect of the instrument measures how much more a one-percentage-point increase in upstream sulfur affects the likelihood of violations after ARP Phase I relative to before. One additional percent of coal sulfur upstream after 1995 leads to a 1.5 to 2.4 percentage point reduction in the likelihood that a utility experiences an MCL or MR violation for nitrates, arsenic, or IOCs of any kind after 1995 compared to before 1995. Since coal production fell most acutely in high sulfur coal areas after 1995, this suggests that in areas where the number of coal mines decreases and upstream contamination subsequently improves, the likelihood of a violation decreases.

In the IV regression, one additional active coal mine increases MCL or MR violations for nitrates, arsenic, and IOCs by 9.9, 7.5, and 6.3 percentage points, respectively. These effects are statistically significant at the 10% level for IOCs, and at the 5% and 1% levels for arsenic and nitrates, respectively. The OLS results underestimate the effect of coal mines on violations relative to the IV results. In general, the OLS impact of coal mines is four to five times smaller than the IV coefficient, suggesting that the omitted variable bias is negative.

Coal mining raises MR violations, supporting the theoretical model's Proposition 2. The results presented in Table 8 estimate equation 5 on MR violations of arsenic, nitrate, and IOCs. An additional coal mine increases the likelihood of an MR violation for nitrates, arsenic, and IOCs by 9.9, 7.4, and 7.1 percentage points, respectively, comprising the entire effect observed when the dependent variable combines MR and MCL violations. These effects are of the same magnitude as the effects of coal mining on a combined measure of MR and MCL violations, so the entire effect of mining on utilities is driven by MR violations rather than MCL violations. The effect of mining on MR violations for IOCs increases in precision compared to combined MR and MCL violations for IOCs. Violations are more than twice as likely to occur relative to the average rate of MR violations after an additional mine operates upstream. Table 1 reports the mean violation rates of arsenic, nitrates, and IOCs at 4 percent, 6.1 percent, and 4.6 percent, respectively.

Coal mining has no detectable effect on the incidence of MCL violations. The impact of coal mining on utility MCL violations for nitrates, arsenic, and IOCs is presented in Table 9. The IV estimated effects of coal mines on the likelihood that a utility experiences a nitrate, arsenic, or inorganic chemical MCL violation are statistically and economically insignificant. The point estimate for the effect of an additional mine on the likelihood a utility receives a nitrate MCL violation is -0.02 percentage points, and for an inorganic chemical MCL it is -0.8 percentage points. None of these effects are statistically significant due to the extreme rarity of MCLs in the sample.

Rising MR violations are consistent with Proposition 2 in the theoretical model. Regulators require more self-reporting from utilities downstream of impaired water, raising self-reporting costs. The theoretical model predicts that mining raises under-reporting through a second channel by changing the compliance costs. Compliance costs rise with mining since the baseline water contamination and the effort required to treat water to any given standard increase, raising compliance costs. As coal mining increases, self-reporting costs and compliance costs rise, both of which increase the benefits of under-reporting.

Coal mining would not affect MCL violations if it did not raise contaminants above the MCL in my sample, or if MR violations concealed contaminant exceedances. In the theoretical model, concealed contaminant exceedances are discovered by the regulator through inspections of under-reporting utilities.

8 The impact of coal mining on regulator visits and enforcement

I estimate the response of regulator visits to coal mining, including sanitary surveys, enforcement visits, and inspections. Table 14 presents the response of regulator visits to coal mining by estimating OLS, equation 5, and the reduced form equations. Column one reports the response of sanitary visits. I find that an additional coal mine leads to a utility experiencing a 14 percentage point increase in sanitary visits by regulators. This is statistically significant at the 1 percent level, and since it represents a 100% increase in sanitary visits over baseline sanitary visit rates, it is economically significant. Regulators are likely to discover concealed contaminant exceedances during sanitary visits and issue punitive or corrective enforcement. Column three reports the response of enforcement visits to coal mining. Enforcement visits increase by 3.7 percentage points. The effect is economically significant, representing a 300% increase over the baseline enforcement visit rate, but it is only significant at the 10% level.

I estimate the effect of coal mining on the enforcement utilities receive. This provides evidence of the results of regulator inspections since regulators would punish utilities harshly for under-reporting in order to conceal contaminant exceedances. This also provides evidence for the disincentives regulators create for utility under-reporting. Table 13 presents the responses of informal, formal, and any type of enforcement to coal mining by estimating OLS, equation 5, and a reduced form. Coal mining reduces the likelihood a utility receives formal enforcement by 5.7 percentage points. This is slightly over 150% of the baseline rate of formal enforcement and significant at the 1% level. This runs counter to the effect of coal mining on enforcement visits. Formal enforcement captures the costlier end of the regulator's toolkit: court orders, penalties, formal letters, compliance orders, and formal notifications. An enforcement visit, by contrast, imposes no direct monetary cost on the utility, such as the cost of installing treatment machinery.

Coal mining leads to less enforcement after taking into account the difference in the likelihood of enforcement visits and formal enforcement. This evidence suggests that coal mining is unlikely to cause utilities to under-report in order to conceal MCL violations, since despite an increase in sanitary visits, formal enforcement falls. At the high rate of sanitary visits caused by increased mining, concealed contaminant exceedances would likely be discovered, and enforcement would be strict; instead, strict formal enforcement falls. The low formal enforcement also suggests that the penalties for under-reporting are relatively mild, indicating that utilities have a low regulatory disincentive to under-report. A low penalty for under-reporting, t in the theoretical model, lowers the self-reporting cutoff $\hat{c}(a)$ at every level of negligence, whatever the probability of inspection and the sanction for a concealed contaminant exceedance.

9 Discussion

By raising self-reporting costs, coal mining can lower self-reporting even if it does not push contamination above the MCL. Contaminant concentration readings do not rise above the MCL across utilities in the SYR2. These utilities differ in their compliance costs, and, returning to the model, in their negligence. This implies that over the range of utilities in the data, the likelihood of exceeding the MCL upon self-reporting is small, $q(a) \to \epsilon$. The self-reporting cutoff $\hat{c}(a) = t - (r - ps)q(a)$ then sits close to t, and a utility deciding whether to report weighs the direct cost of self-reporting against the penalty for skipping it, without a contamination exceedance to conceal if they were to self-report.

I test the effect of self-reporting costs on self-reporting by estimating the effect of increased nitrates testing requirements on the likelihood that a utility under-reports. Nitrates, unlike other IOCs, require increased testing when contaminant concentrations exceed 50% of the nitrates MCL. Other IOCs increase testing based on source water risk instead of a single contaminant concentration threshold. I estimate the effect of SYR2 contaminant concentrations of nitrates on the likelihood of nitrate self-reporting violations. Table 12 presents the results; columns one and two report the likelihood of an MR violation for nitrates within one year and within six months of a contaminant reading. A nitrate reading above 50% of the nitrate MCL increases the likelihood of a nitrate self-reporting violation by 59 percentage points and 26 percentage points within the next twelve and six months, respectively. I find no effect on the likelihood of an MR violation for nitrates in the twelve or six months following a one standard deviation increase in mean nitrate readings. These effects are suggestive and not causal.

I find evidence that under-reporting receives low enforcement: net enforcement falls as mining increases and self-reporting decreases. This reduces the disincentive to under-report due to cost increases associated with self-reporting. The result is that regulators are required to increase sanitary visits to utilities to determine the water quality and, ultimately, the health risk of an under-reporting utility. The welfare effects of this are not a priori clear. The regulators may be able to monitor with economies of scale that are lower than the utilities' cost of self-reporting. However, the economies of scale would have to overcome the large amount of resources that the regulator would have to devote to monitoring all utilities in a timely manner.

The combined evidence that coal mining causes regulators to conduct more sanitary visits while issuing less enforcement suggests that utilities are not concealing contaminant exceedances. In addition, SYR2 contaminant concentrations are far below the MCL. A utility would not exceed the MCL for barium, nitrates, or selenium, even at the maximum contaminant level observed in the sample and the highest single-year dose of upstream coal production: 10 million short tons. Arsenic is the exception: the sample maximum of 0.0110 mg/L stays below the pre-2006 MCL of 0.050 mg/L but exceeds the stricter 0.010 mg/L MCL that took effect in 2006. This suggests that utilities maintain water quality well below MCL levels. However, this result is limited to a subset of utilities and a subset of the time period included in SYR2. Sanitary visits at the high levels seen in response to coal mining would fail to reveal contamination violations because utilities were not violating contamination limits. If utilities were concealing exceedances, the rise in sanitary visits and enforcement visits should have converted into more enforcement; instead, formal enforcement falls, and the concentration readings that are available sit well below the contaminant limit. I therefore interpret the fall in self-reporting as a response to the rising cost of self-reporting rather than as the concealment of a contaminant exceedance, subject to the concentration data covering only part of the sample. Monitoring gaps of this kind can be overcome by continuous monitoring technology that the regulator controls, making violations easier to spot and successful enforcement more likely.

10 Conclusion

I present evidence that under a self-reporting regulatory framework designed to monitor a quality target, self-reporting falls if contamination increases, even if that quality does not fall below regulatory standards. I show this in the context of coal mining and utilities: utility SDWA self-reporting of water quality falls as water quality exogenously deteriorates. An additional active coal mine upstream raises the likelihood of a monitoring and reporting violation for inorganic chemicals, arsenic, and nitrates by seven to ten percentage points, while leaving the likelihood of a contaminant limit violation unchanged. Utility SDWA self-reporting becomes costlier as coal mining increases since an increase in coal mining leads to increases in IOC water contamination at utilities, which raises the cost of self-monitoring. Consistent with that cost channel, a nitrate reading above half the contaminant limit, the level at which the regulator requires more frequent testing, is followed by a large increase in the likelihood of a nitrate self-reporting violation. I do not detect contaminant concentration levels increasing above maximum contaminant levels and cannot conclude that upstream coal mining is increasing pollution to the extent that utilities need to install improved abatement technology to meet SDWA maximum contaminant level compliance.

I find that penalties for not self-reporting are not high enough to induce utilities to self-report as self-reporting costs rise. Regulators respond to upstream mining with attention rather than sanction: an additional upstream mine raises the likelihood of a sanitary visit by 14 percentage points and of an enforcement visit by 3.7 percentage points, while lowering the likelihood of formal enforcement by 5.7 percentage points. That regulators visit these utilities so much more often without formal enforcement following suggests the visits find nothing to sanction, and that utilities are not under-reporting in order to conceal a contaminant exceedance. The regulator can use sanitary visits to assess the threat posed to public health by an MR violation and can implement a less costly type of enforcement when the threat is discovered to be low. However, without punishing non-self-reporting at high levels, regulators will have to conduct more sanitary visits to ensure that contaminant concentration levels do not exceed health limits.

I do not address the efficiency of shifting monitoring onto the regulators. Water quality is generally perceived as a local issue with limited externalities (Olmstead 2010). Without externalities, it may be inefficient for a state regulator to expend resources to monitor a local utility. However, regulators may be able to conduct monitoring at a lower marginal cost than small utilities. Small utilities find the burden of water quality management particularly heavy. I would need to take into account the regulators' preferences in order to measure efficiency. Regulator preferences affect enforcement, not only violation severity (Kang and Silveira 2021).

The findings here suggest that the regulatory design of the SDWA, which relies on utility self-monitoring and assigns substantially lower penalties to non-monitoring than to detected contamination, creates systematic incentives to under-report. Increasing penalties for non-monitoring in high-pollution areas may be a more effective lever for improving drinking water outcomes than increases in MCL stringency alone.

A Proofs of the propositions

This appendix contains the derivation of the utility's negligence choice and the statements and proofs of the three propositions summarized in Section 3. Notation follows the setup there.

A.1 Optimal negligence

The utility's problem is to minimize the losses of the opportunity cost of choosing a negligence below $\bar{a}$ and the expected cost $e(a;c)$. Therefore, the utility maximizes $-\theta[b(\bar{a}) - b(a)] - e(a;c)$ with respect to a. The first-order condition is piecewise:

$$\theta b'(a) = r\, q'(a) \qquad \text{(self-reporting branch)} \tag{6}$$

$$\theta b'(a) = ps\, q'(a) \qquad \text{(under-reporting (MR) branch)} \tag{7}$$

Because $r > ps$, the marginal penalty is lower on the under-reporting branch than on the self-reporting branch.

Which branch a utility takes is not settled by $\hat{c}(a)$ evaluated at either candidate negligence level, because $\hat{c}$ is itself decreasing in the negligence the utility goes on to choose. Proposition 2 shows that the two branch payoffs cross at a single cost draw, and that this crossing point lies between $\hat{c}$ evaluated at the two candidates. Utilities that draw a higher self-reporting cost are more likely to under-report, and a rightward shift in the population's cost distribution raises the share of utilities that do so without moving any type's own crossing point.

A.2 The propositions

The model predicts that when water quality deteriorates due to increased contamination, self-reporting rates fall.

Proposition 1: Negligence is increasing in compliance costs. Within each reporting branch,

$$\frac{\partial a}{\partial \theta} = \frac{b'(a)}{-\theta b''(a) + \rho\, q''(a)} > 0,$$

where $\rho \in \{r, ps\}$ is the branch's sanction. Under the interiority condition the first-order condition characterizes the optimum, and the denominator is positive because $b'' < 0$ and $q'' \geq 0$, which is also the second-order condition for the negligence problem within a branch. Since $r > ps$, a utility that switches from self-reporting to under-reporting faces a lower marginal sanction and moves to a higher negligence level, so negligence rises with θ overall, with an upward jump at the switch.

Proposition 2: The share of under-reporting utilities increases as contamination increases, $\partial MR/\partial m \geq 0$.

For a compliance-cost type θ facing a marginal sanction $\rho > 0$, let $a(\rho;\theta)$ denote the negligence level solving $\theta b'(a) = \rho\, q'(a)$. The negligence a self-reporting utility chooses is $a_{SR}(\theta) \equiv a(r;\theta)$ and the negligence an under-reporting utility chooses is $a_{MR}(\theta) \equiv a(ps;\theta)$, with $a_{SR}(\theta) < a_{MR}(\theta)$ since $r > ps$.

The utility's objective is not globally concave: the marginal penalty drops from $r\,q'(a)$ to $ps\,q'(a)$ at the negligence level where $\hat{c}(a) = c$. Writing the utility's problem as a maximization, $\max_a\{\theta b(a) - \theta b(\bar{a}) - \min[\, c + rq(a),\;\; t + ps\,q(a)\,]\}$, the inner minimum enters with a negative sign and is therefore a maximum over the two reporting branches, which can be exchanged with the maximization over a: $\max_a \max\{\cdot,\cdot\} = \max\{\max_a(\cdot),\, \max_a(\cdot)\}$, since neither branch's maximizer depends on the other's value. So the value of the problem is the larger of two per-branch maxima, each built from $W(\rho;\theta) \equiv \max_{a\in[0,\bar{a}]}\{\theta b(a) - \rho\, q(a)\}$ for $\rho \in \{r, ps\}$, each attained at a unique $a(\rho;\theta)$ since $\theta b(a) - \rho\, q(a)$ is strictly concave. No negligence level besides $a_{SR}(\theta) \equiv a(r;\theta)$ and $a_{MR}(\theta) \equiv a(ps;\theta)$ is ever optimal, for any c. The payoff from self-reporting is $W(r;\theta) - \theta b(\bar{a}) - c$ and the payoff from under-reporting is $W(ps;\theta) - \theta b(\bar{a}) - t$, and the utility takes whichever is larger. The former falls one-for-one in c and the latter does not involve c at all, so their difference is affine in c with nonzero slope and crosses zero exactly once. Cancelling $\theta b(\bar{a})$ and applying the envelope theorem, $\partial W(\rho;\theta)/\partial\rho = -q(a(\rho;\theta))$, the crossing point is the type specific switching cost

$$c^*(\theta) \equiv t + W(r;\theta) - W(ps;\theta) = t - \int_{ps}^{r} q\big(a(\rho;\theta)\big)\, d\rho. \tag{8}$$

Since $q(a(\rho;\theta)) \leq q(\bar{a})$ for every ρ, the maintained assumption $t \geq (r - ps)q(\bar{a})$ delivers $c^*(\theta) \geq 0$. When negligence carries no exceedance risk, $q \equiv 0$, the integral vanishes and the cutoff is simply the under-reporting penalty t. The cutoff depends only on the sanctions and on the utility's own compliance-cost type, not on contamination m.

A type-θ utility under-reports if and only if its cost draw exceeds this cutoff, $c > c^*(\theta)$, so the share of utilities under-reporting and receiving an MR violation is

$$MR(m) = \int_0^{\bar{\theta}} \left[\, 1 - G(c^*(\theta); m)\,\right] f(\theta; m)\, d\theta \;\equiv\; \int_0^{\bar{\theta}} h(\theta, m)\, f(\theta; m)\, d\theta. \tag{9}$$

Differentiating (9) splits into two channels:

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m)\, d\theta}_{\text{Channel A: cost-distribution shift}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m}\, d\theta}_{\text{Channel B: type-distribution shift}} . \tag{10}$$

Channel A:

$$\frac{\partial h(\theta, m)}{\partial m} = -\frac{\partial G(c^*(\theta); m)}{\partial m} \geq 0 \quad \text{for every } \theta,$$

since $\partial G(\cdot\,; m)/\partial m \leq 0$ pointwise.

Channel A is therefore non-negative: holding the type composition fixed, a rightward shift in the self-reporting cost distribution pushes more mass of c above each type's unchanged cutoff.

Channel B:

The maintained assumption signs $\partial F(\theta; m)/\partial m \leq 0$, but Channel B is written in terms of the *density* derivative $\partial f(\theta; m)/\partial m$, which has no assumed sign — a rightward shift in a cdf can raise the density in one part of the support and lower it in another. To use the assumption we actually have, rewrite Channel B in terms of F instead of f via integration by parts.

Define $\varphi(\theta; m) \equiv \partial F(\theta; m)/\partial m$. Since $f(\theta; m) = \partial F(\theta; m)/\partial \theta$, swapping the order of the two partial derivatives gives $\partial f(\theta; m)/\partial m = \partial \varphi(\theta; m)/\partial \theta$, so Channel B is

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial \varphi(\theta; m)}{\partial \theta}\, d\theta.$$

Integrating by parts (treating $[\partial \varphi(\theta; m)/\partial \theta]\, d\theta$ as $d\varphi$),

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial \varphi(\theta; m)}{\partial \theta}\, d\theta = \Big[h(\theta, m)\, \varphi(\theta; m) \Big]_0^{\bar{\theta}} - \int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial \theta}\, \varphi(\theta; m)\, d\theta.$$

The boundary term vanishes: $\varphi(0; m) = \partial F(0; m)/\partial m = 0$ and $\varphi(\bar{\theta}; m) = \partial F(\bar{\theta}; m)/\partial m = 0$, since $F(0; m) = 0$ and $F(\bar{\theta}; m) = 1$ for every m (the cdf is pinned at the endpoints of its support regardless of m). Substituting $\varphi(\theta; m) = \partial F(\theta; m)/\partial m$ back in,

$$\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m}\, d\theta = -\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial \theta} \frac{\partial F(\theta; m)}{\partial m}\, d\theta. \tag{11}$$

Signing (11) therefore requires the slope of h in θ. Differentiating (8) and applying the envelope theorem in θ, $\partial W(\rho; \theta)/\partial \theta = b(a(\rho; \theta))$,

$$c^{*\prime}(\theta) = b(a_{SR}(\theta)) - b(a_{MR}(\theta)) < 0,$$

since $a_{SR}(\theta) < a_{MR}(\theta)$ and $b' > 0$. Then

$$\frac{\partial h(\theta, m)}{\partial \theta} = -g(c^*(\theta); m)\, c^{*\prime}(\theta) \geq 0,$$

so h is non-decreasing in θ: higher-cost types choose more negligence, which lowers their own switching cutoff and makes them more likely to under-report.

$$\text{Channel B} = -\int_0^{\bar{\theta}} \underbrace{\frac{\partial h(\theta, m)}{\partial \theta}}_{\geq 0} \underbrace{\frac{\partial F(\theta; m)}{\partial m}}_{\leq 0} d\theta \;\geq\; 0.$$

Both channels are therefore non-negative, so $\partial MR/\partial m$ = Channel A + Channel B ≥ 0.

Contamination from mining can raise MR violations by increasing self-reporting costs relative to each type's fixed cutoff and by shifting the population toward higher-cost types who are already structurally closer to their own cutoff.

Proposition 3: Increasing all penalties increases the share of utilities self-reporting.

Consider scaling every sanction by a common factor $\lambda > 0$: $(r, s, t) \to \lambda(r, s, t)$, holding $q(\cdot)$, $b(\cdot)$, p, and the cost draws θ, c fixed. Costs become

$$\begin{aligned} \text{self-report:} \quad & c + q(a)\, \lambda r \\ \text{under-report (MR):} \quad & \lambda t + p\, [q(a)\, \lambda s] \end{aligned}$$

Differentiating the expected annual cost in a on each branch,

$$\begin{aligned} \frac{\partial e_\lambda(a; c)}{\partial a} &= \lambda r\, q'(a) = \lambda\, \frac{\partial e(a; c)}{\partial a} && \text{(self-reporting branch)} \\ \frac{\partial e_\lambda(a; c)}{\partial a} &= \lambda ps\, q'(a) = \lambda\, \frac{\partial e(a; c)}{\partial a} && \text{(under-reporting (MR) branch)} \end{aligned}$$

Negligence falls in λ.

$$\frac{\partial a}{\partial \lambda} = \frac{\rho\, q'(a)}{\theta b''(a) - \lambda \rho\, q''(a)}.$$

The second-order condition invoked in Proposition 1, $\lambda \rho\, q''(a) - \theta b''(a) \geq 0$, makes the denominator non-positive, so $\partial a/\partial \lambda \leq 0$: negligence is weakly decreasing in the penalty scale, within each branch.

The reporting rule scales by λ. The self-report condition with the scaled penalties, $c + \lambda r\, q(a) \leq \lambda[t + ps\, q(a)]$ rearranges to

$$c \leq \hat{c}_\lambda(a) \equiv \lambda\big[\, t - (r - ps)\, q(a)\,\big] = \lambda\, \hat{c}(a). \tag{12}$$

Unlike $\hat{c}_\lambda(a)$ itself, the self-report cost c does not scale with λ since it is a technology/labor cost, not a legal penalty, so scaling λ up moves the threshold $\hat{c}_\lambda(a)$ relative to a fixed distribution of c.

Scaling the sanctions replaces the marginal sanction pair (r, ps) with $(\lambda r, \lambda ps)$ and the under-reporting penalty t with λt, so the switching cost of (8) becomes

$$C^*(\lambda;\theta) = \lambda t - \int_{\lambda ps}^{\lambda r} q(a(u;\theta))\, du,$$

with derivative

$$\frac{\partial C^*(\lambda;\theta)}{\partial \lambda} = t - \Big[r\, q(a(\lambda r;\theta)) - ps\, q(a(\lambda ps;\theta)) \Big] \;\geq\; t - (r - ps)\, q(\bar{a}) \;\geq\; 0.$$

The first inequality uses $a(\lambda r;\theta) \leq a(\lambda ps;\theta)$, so that $q(a(\lambda r;\theta)) \leq q(a(\lambda ps;\theta))$ and the bracketed term is at most $(r - ps)\, q(a(\lambda r;\theta)) \leq (r - ps)\, q(\bar{a})$. The second is the maintained assumption on t. Raising the penalty scale therefore raises the switching cost of every type, holding the population's cost draws fixed. Aggregating over types, the share of utilities that self-report,

$$SR(\lambda) = \int_0^{\bar{\theta}} G(C^*(\lambda;\theta); m)\, f(\theta; m)\, d\theta,$$

is non-decreasing in λ, since G is increasing in its first argument and $\partial C^*(\lambda;\theta)/\partial\lambda \geq 0$ for every θ. A higher penalty scale reduces negligence and increases self-reporting.

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Tables and Figures

Table 1: Coal Mining Exposed Utilities' Inorganic Chemical Water Violations 1985–2005

Panel A: Any violation in year

	MR		MCL	
Contaminant	**%**	**N**	**%**	**N**
Nitrates	6.05	377	0.06	4
Arsenic	4.01	250	0.03	2
IOCs	4.59	286	0.32	20

Panel B: Days in year

	MR				MCL			
Contaminant	**Mean**	**SD**	**P90**	**P99**	**Mean**	**SD**	**P90**	**P99**
Nitrates	15.53	67.32	0.00	365.00	0.18	8.01	0.00	0.00
Arsenic	10.69	56.01	0.00	365.00	0.04	2.59	0.00	0.00
IOCs	12.27	60.15	0.00	365.00	0.78	15.83	0.00	0.00

Notes: Sample of drinking water utilities downstream of a coal mine between 1985–2005. MR = monitoring and reporting violation; MCL = maximum contaminant level violation. Panel A: % Non-zero is the share of utility-year observations with a nonzero violation share for that category, in percent; Num. Violations is the corresponding count of utility-year observations. Panel B: Mean, SD, P90, and P99 describe the number of days in a year in violation. Number of observations = Number of utilities $\times$ Number of years. N = 6,232 = 340 utilities $\times$ up to 21 years (1985–2005).

Table 2: Enforcement Actions and Site Visits, Coal Mining Exposed Utilities 1985–2005

	Whole panel		During MR year		During MCL year	
	%	N	%	N	%	N
Panel A: Enforcement type						
Formal	3.50	218	9.53	43	7.69	2
Informal	19.02	1,184	67.63	305	53.85	14
Any	21.37	1,330	70.73	319	65.38	17
Panel B: Visit type						
Sanitary	14.55	906	14.19	64	7.69	2
Technical assistance	2.33	145	4.88	22	0.00	0
Enforcement	1.16	72	2.44	11	3.85	1
Sample collection	0.90	56	0.44	2	0.00	0
Inspection	4.05	252	5.76	26	3.85	1

Notes: Sample of drinking water utilities downstream of a coal mine between 1985–2005. MR = monitoring and reporting violation; MCL = maximum contaminant level violation. Whole panel columns report the share and count of utility-year cells with a given enforcement action or site visit type, among all 6,225 cells. During MR year and During MCL year columns restrict to the 451 and 26 cells, respectively, with a nitrate, arsenic, or inorganic chemical violation of that category, before computing the same share and count. Panel A: Formal and Informal enforcement follow the EPA SDWIS enforcement-action classification; Any is a cell with an enforcement action of either category. Panel B: Sanitary visits are sanitary surveys and follow-up sanitary surveys; Technical assistance covers technical assistance, engineering, and operations and maintenance visits; Enforcement visits are formal-enforcement, investigation, and emergency site visits; Sample collection is a sample-collection visit; Inspection covers site, regularly scheduled, and informal system inspections. An enforcement action or visit type may co-occur with others in the same cell, so rows need not sum to the Any/whole-panel total. Number of observations = 6,225.

Table 3: Utility contaminant concentrations (1998-2005) and cumulative upstream coal production exposure (since 1985)

Variable	MCL	Mean	Max	Std. Dev.	N	Near MCL
Panel A: Full sample						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0029	0.0110	0.0023	378	0.0%
Nitrate (mg/L)	10.0 mg/L	0.7520	5.5400	0.8482	444	0.5%
Barium (mg/L)	2.000 mg/L	0.0748	0.7071	0.0671	369	0.0%
Selenium (mg/L)	0.050 mg/L	0.0048	0.0354	0.0051	367	0.5%
Cumul. upstream coal prod. (10M ST)	—	0.4402	14.1792	1.4162	518	—
Panel B: Utility exposed to above median cumulative tons of coal extraction						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0031	0.0100	0.0026	177	0.0%
Nitrate (mg/L)	10.0 mg/L	0.7770	5.5400	0.8175	236	0.4%
Barium (mg/L)	2.000 mg/L	0.0754	0.3320	0.0579	169	0.0%
Selenium (mg/L)	0.050 mg/L	0.0040	0.0346	0.0043	169	0.6%
Cumul. upstream coal prod. (10M ST)	—	0.8822	14.1792	1.9089	258	—

Notes: SYR2 sample from 1998–2005. Cumulative upstream coal production (10M ST) is cumulative coal production since 1985, in units of 10 million short tons, in the watershed immediately upstream of the utility's intake. For cumulative upstream coal production, statistics are computed over unique utility$\times$year pairs that appear in at least one regression sample. "Near MCL" is the share of utility-year mean concentrations exceeding 50% of the applicable MCL. Panel B restricts Panel A's sample to utility-year observations with cumulative upstream coal production above the sample median of 0.0094 (10M ST).

Table 4: MR and MCL violation rates and counts by SYR2 contaminant reporting status

Panel A: 1997–2005

	Utility appears in SYR2			
	Yes	No	Diff.	p-value
	(1)	(2)	(3)	(4)
Ever MR violation	32.4%	9.8%	22.6%***	0.000
Ever MCL violation	1.3%	0.0%	1.3%*	0.083
Mean annual violations (count, 0–6)	0.09	0.11	-0.02	0.663

Panel B: 1985–2005

	Utility appears in SYR2			
	Yes	No	Diff.	p-value
	(1)	(2)	(3)	(4)
Ever MR violation	51.1%	32.8%	18.3%***	0.001
Ever MCL violation	5.3%	1.6%	3.7%*	0.052
Mean annual violations (count, 0–6)	0.13	0.26	-0.13**	0.018

Notes: Sample restricted to utilities strictly downstream of a coal mine. *Has SYR2 reading*: at least one Six-Year Review contaminant reading for arsenic, nitrate, selenium, or barium. *No SYR2 reading*: utility is in a state where at least one utility has a SYR2 reading but the utility itself does not. MR violation: IOC monitoring and reporting violation. MCL violation: IOC maximum contaminant violation. Mean annual violations: sum of MR and MCL violation indicators across arsenic, nitrate, and inorganic chemicals categories (maximum of 6 per utility-year). N utilities: group 1 = 225, group 2 = 122. Diff. stars from two-sample t-test: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: First stage: effect of the Acid Rain Program on the number of upstream coal mines (summed across upstream watersheds, utilities at most one HUC12 downstream)

	Upstream coal mines (sum)
	(1)
post95 × Upstream sulfur %	-0.2404***
	(0.0458)
F-test (1st stage, clustered)	27.52
Observations	6,225

Notes: The dependent variable is the number of coal mines in watersheds upstream of the utility's intake, summed across upstream watersheds. The instrument interacts an indicator for the post-1995 period with the sum of coal sulfur content across upstream watersheds. The sample is utilities at most one watershed downstream of a coal mine. All specifications include utility and year fixed effects. Standard errors clustered at the utility level. Sample period 1985–2005. *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Effect of instrument on utility characteristics

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.0384
		(0.0735)
Post-1995 × Upstream sulfur	-0.0016	-0.0016
	(0.0016)	(0.0017)
Utility fixed effects	✓	
Year fixed effects	✓	✓
Balanced panel		✓
Utilities	333	260
Observations	6,225	5,460

Notes: Standard errors, clustered at the utility level, are shown in parentheses below each coefficient. Sample period 1985–2005. *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Effect of coal mines on inorganic chemical violations at utilities

	Any violation		
	Nitrates	Arsenic	Inorganic chemicals
	(1)	(2)	(3)
OLS			
Upstream coal mines (sum)	1.94***	1.67**	1.75***
	(0.72)	(0.65)	(0.65)
2SLS			
Upstream coal mines (sum)	9.91***	7.50**	6.34*
	(3.72)	(3.06)	(3.32)
RF			
post95 $\times$ Upstream sulfur %	-2.38***	-1.80**	-1.52*
	(0.84)	(0.73)	(0.80)

Notes: Dependent variable equals 1 if the utility had any violation of that type during the year, 0 otherwise; coefficients and standard errors multiplied by 100 to show percentage point change. The instrument interacts an indicator for the post-1995 period with the sum of coal sulfur content across upstream watersheds. All specifications include utilities and year fixed effects. Standard errors clustered at the utility level. The first-stage F-statistic (clustered at the utility level) for the instrumented variable, Upstream coal mines (sum), is 27.52. Sample period 1985–2005. Number of observations = 6,225. *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Effect of coal mines on inorganic chemical monitoring and reporting violations at utilities

	Monitoring and reporting (MR) violation		
	Nitrates	Arsenic	Inorganic chemicals
	(1)	(2)	(3)
OLS			
Upstream coal mines (sum)	1.95***	1.70***	1.80***
	(0.72)	(0.65)	(0.66)
2SLS			
Upstream coal mines (sum)	9.93***	7.43**	7.07**
	(3.73)	(3.05)	(3.23)
RF			
post95 $\times$ Upstream sulfur %	-2.39***	-1.79**	-1.70**
	(0.84)	(0.73)	(0.77)

Notes: Dependent variable equals 1 if the utility had any violation of that type during the year, 0 otherwise; coefficients and standard errors multiplied by 100 to show percentage point change. The instrument interacts an indicator for the post-1995 period with the sum of coal sulfur content across upstream watersheds. All specifications include utilities and year fixed effects. Standard errors clustered at the utility level. The first-stage F-statistic (clustered at the utility level) for the instrumented variable, Upstream coal mines (sum), is 27.52. Sample period 1985–2005. Number of observations = 6,225. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 9: Effect of coal mines on inorganic chemical maximum contaminant level violations at utilities

	Maximum contaminant level (MCL) violation		
	Nitrates	Arsenic	Inorganic chemicals
	(1)	(2)	(3)
OLS			
Upstream coal mines (sum)	-0.02	-0.02	-0.00
	(0.02)	(0.04)	(0.14)
2SLS			
Upstream coal mines (sum)	-0.02	0.08	-0.78
	(0.02)	(0.06)	(0.69)
RF			
post95 $\times$ Upstream sulfur %	0.00	-0.02	0.19
	(0.00)	(0.01)	(0.16)

Notes: Dependent variable equals 1 if the utility had any violation of that type during the year, 0 otherwise; coefficients and standard errors multiplied by 100 to show percentage point change. The instrument interacts an indicator for the post-1995 period with the sum of coal sulfur content across upstream watersheds. All specifications include utilities and year fixed effects. Standard errors clustered at the utility level. The first-stage F-statistic (clustered at the utility level) for the instrumented variable, Upstream coal mines (sum), is 27.52. Sample period 1985–2005. Number of observations = 6,225. *** p$<$0.01, ** p$<$0.05, * p$<$0.1.

Table 10: Balance test of utility characteristics and upstream coal production 1998–2005

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.0601		-0.1413	
	(0.1302)		(0.1077)	
Log backcast pop. 1997	0.0473	0.0519	0.0105	0.0107
	(0.0425)	(0.0445)	(0.0104)	(0.0109)
Log pop. served (SYR2)	-0.0489	-0.0373	-0.0062	-0.0024
	(0.0418)	(0.0319)	(0.0219)	(0.0230)
Num. intake facilities	0.0600	0.0318	0.0953***	0.0932***
	(0.0429)	(0.0328)	(0.0332)	(0.0351)
Surface water source	0.0820	-0.1104	0.1663	0.1773
	(0.0849)	(0.1655)	(0.1159)	(0.1271)
Public ownership	-0.0954	-0.0404	0.1426*	0.1561*
	(0.1584)	(0.1204)	(0.0756)	(0.0897)
Wholesaler	-0.1556	-0.3919	-0.2090	-0.2173
	(0.1173)	(0.4044)	(0.1286)	(0.1364)
Joint F-test (all covariates)	1.21	0.58	5.67	4.01
p-value	0.308	0.748	0.000	0.001
Observations	122	122	122	122
HUC02 fixed effects		✓		✓

Notes: Covariates differ in timing, and only population pre-dates the dose window. Backcast population served in 1997 is constructed from decennial census geography and interpolated between the 1990 and 2000 census anchors; it is therefore dated before 1998 by construction but is not an observed 1997 measurement. The remaining covariates (population served, number of intake facilities, number of source HUC12s, primary source type, ownership, wholesaler status, and source water protection) are drawn from the SDWIS/SYR2 inventory and are recorded contemporaneously with the dose window. Heteroskedasticity-robust standard errors. Sample: utilities downstream of a coal mine 1998–2005. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Effect of cumulative upstream coal production on Inorganic Chemicals, SYR2 1998–2005

	Mean conc.			
	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Cumul. upstream coal prod. (10M ST)	0.0023***	0.0572	0.0171*	0.0033**
	(0.0004)	(0.0922)	(0.0093)	(0.0016)
Observations	334	432	322	319
Utility fixed effects	✓	✓	✓	✓
huc02-year fixed effects	✓	✓	✓	✓

Notes: Standard errors clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

Table 12: Nitrate MR violations following a reading above 50% of the MCL (downstream-of-mine sample)

	Nitrate MR (1-yr)	Nitrate MR (6-mon)
	(1)	(2)
Concen. > 50% MCL	58.9693**	26.1081**
	(24.6040)	(10.5298)
Mean concen. (z-score)	1.2804	0.0482
	(1.4274)	(0.8741)
Observations	851	851

Notes: SYR2 sample restricted to utilities strictly downstream of a coal mine (1998–2005), nitrate only. Outcome: nitrate MR (monitoring/reporting) violation in the forward window (1–365 days for the 1-yr column; 1–182 days for the 6-mon column) following the sample date. Concen. > 50% MCL = reading at 50–100% of the MCL, the quarterly-monitoring trigger. Mean concentration = utility-year mean reading, z-scored within chemical. Coefficients and standard errors are in percentage points. All specifications include utility and year fixed effects. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. SEs clustered at the utility level.

Table 13: Effect of Coal Mining on Enforcement Actions by Type (Downstream Sample)

	Any enforcement		
	Informal	Formal	None
	(1)	(2)	(3)
OLS			
Upstream coal mines (sum)	0.84	-1.69***	-0.78
	(0.79)	(0.45)	(0.86)
2SLS			
Upstream coal mines (sum)	-0.56	-5.65***	-0.10
	(4.00)	(2.05)	(4.23)
RF			
post95 × Upstream sulfur %	0.14	1.45***	0.03
	(1.02)	(0.52)	(1.08)

Notes: Sample restricted to utilities strictly downstream of a coal mine. Each outcome is equal to 100 if the enforcement event occurred in the utility-year and 0 otherwise; coefficients and standard errors are in percentage points. Informal enforcement actions occur in 19.0% of the panel, formal enforcement actions in 3.5% of the panel, and no enforcement in 78.7% of the panel. The instrument interacts an indicator for the post-1995 period with mean upstream coal sulfur content. All specifications include utilities and year fixed effects. SEs clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 14: Effect of Coal Mining on Regulator Visit Probability by Visit Type (Downstream Sample, LPM)

	Any visits				
	Sanitary	Technical assistance	Enforcement	Sample collection	Inspection
	(1)	(2)	(3)	(4)	(5)
OLS					
Upstream coal mines (sum)	1.06	-0.38	-0.71**	-0.19	-0.33
	(0.67)	(0.42)	(0.32)	(0.12)	(0.47)
2SLS					
Upstream coal mines (sum)	14.06***	1.41	3.67*	1.35	-1.88
	(4.87)	(1.96)	(1.88)	(1.15)	(2.41)
RF					
post95 × Upstream sulfur %	- 3.60***	-0.36	-0.94**	-0.34	0.48
	(1.03)	(0.48)	(0.40)	(0.27)	(0.60)

Notes: Sample restricted to utilities strictly downstream of a coal mine. N = 6232 utility-years. Panels (OLS, 2SLS, reduced form) report coefficients for each visit-type outcome shown in the columns, equal to 100 if any visit of that type occurred in the utility-year and 0 otherwise; coefficients and standard errors are in percentage points. Sanitary visits are sanitary surveys and follow-up sanitary surveys. Technical assistance includes technical assistance, engineering determination/advice/plan review, and operation and maintenance visits. Enforcement visits include formal enforcement, investigation, and emergency assistance visits. Sample collection is sample collection visits. Inspection includes site inspections, regularly scheduled visits, and informal system inspections. The instrument interacts an indicator for the post-1995 period with mean upstream coal sulfur content. All specifications include utilities and year fixed effects. SEs clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Figure 1: Hierarchical structure of USGS hydrologic unit codes

Notes: Illustrates how a hydrologic region is nested into progressively smaller watershed units, from two-digit units covering large multi-state regions down to twelve-digit units covering a few square miles. Each smaller unit is fully contained within the boundaries of the larger unit above it.

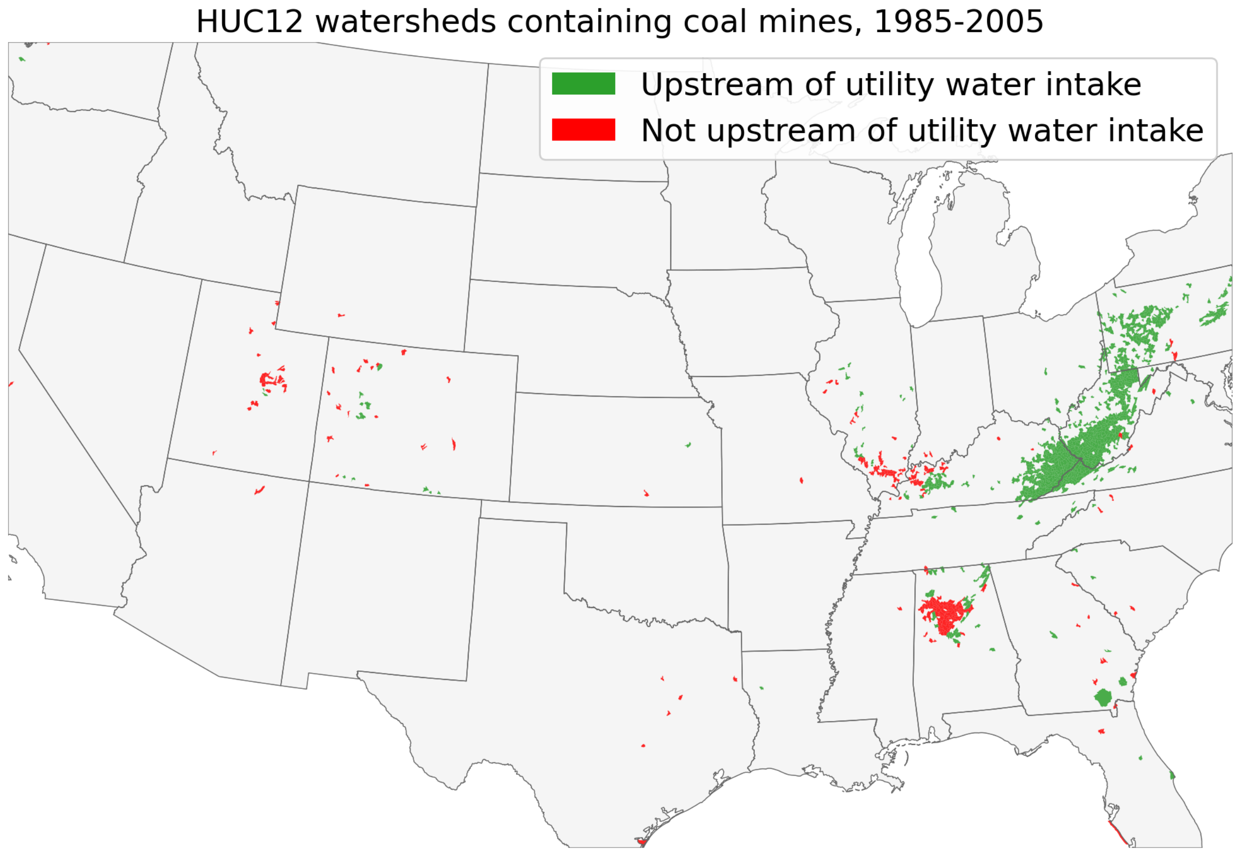

Figure 2: Shading distinguishes HUC12 watersheds that lie directly upstream of a water utility intake and contain a coal mine from HUC12 watersheds containing a coal mine with no water utility located downstream.

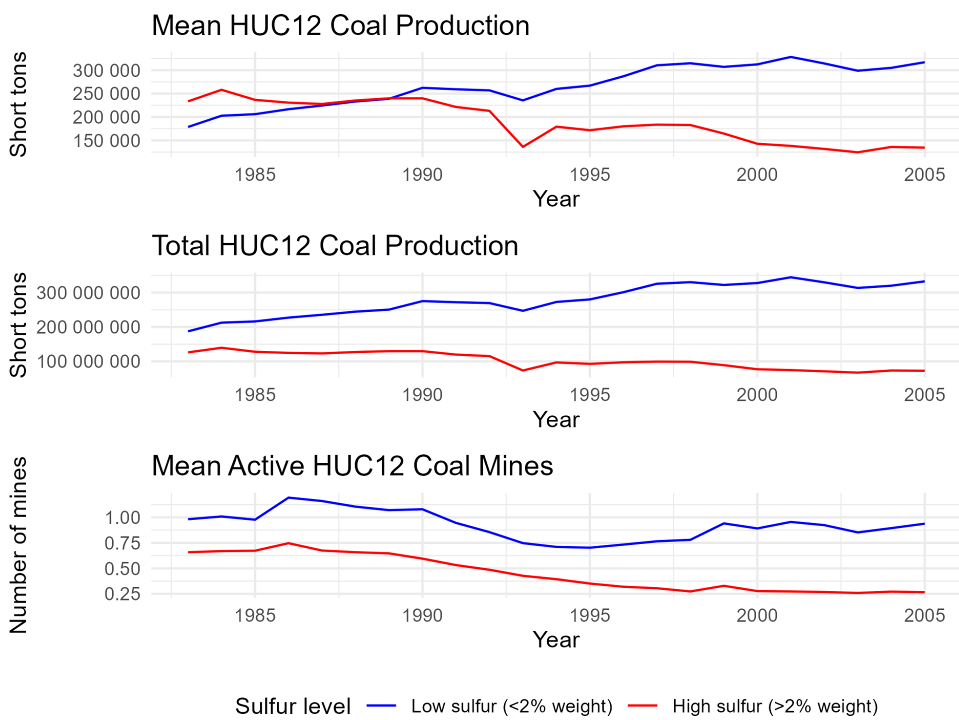

Figure 3: Coal production trends by HUC12 watershed sulfur content, 1983 to 2005.

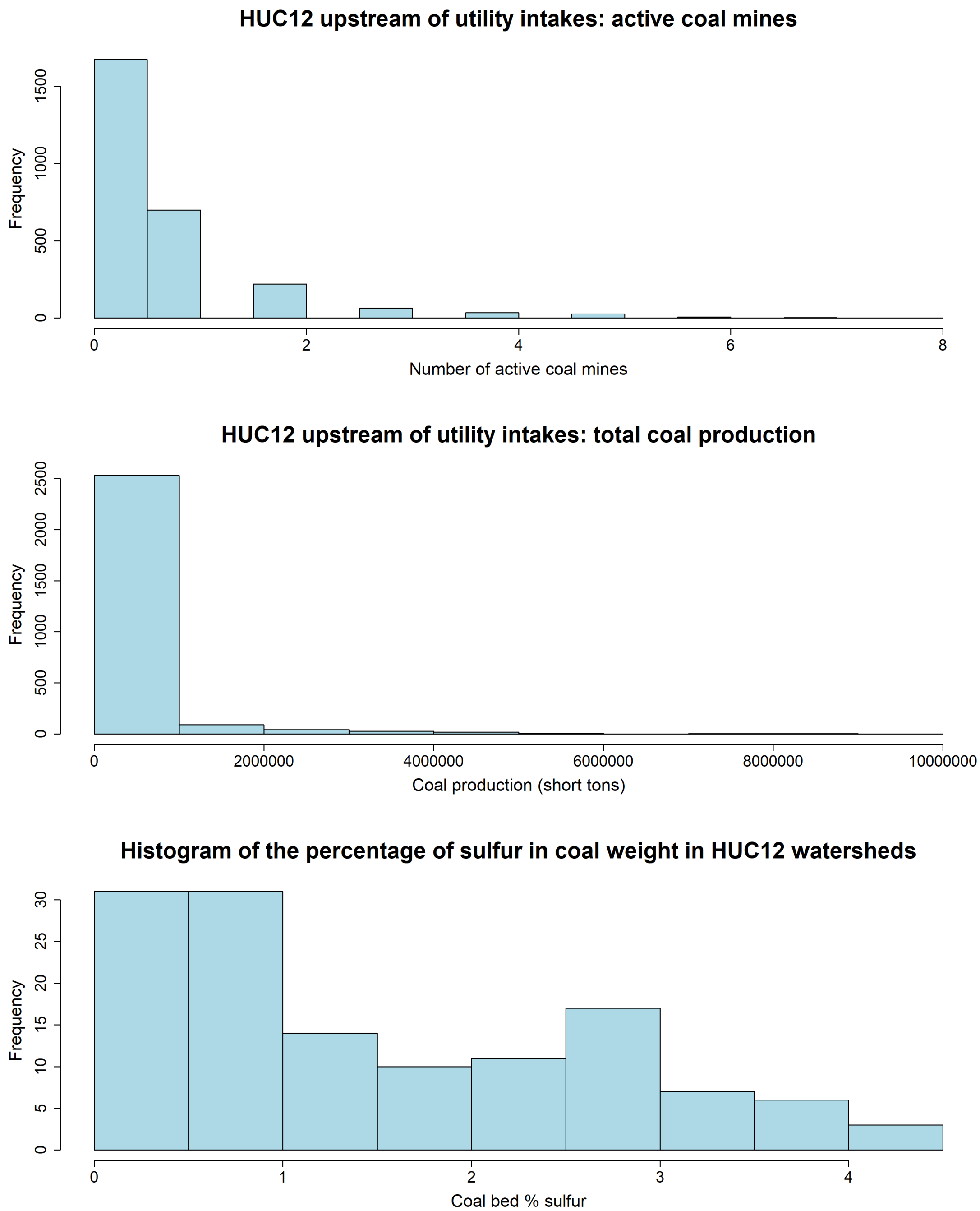

Figure 4: Histogram of the percentage of sulfur in coal weight in HUC12 watersheds downstream of coal production and where a water utility intake is located. The top two panels show histograms of utility-year observations.

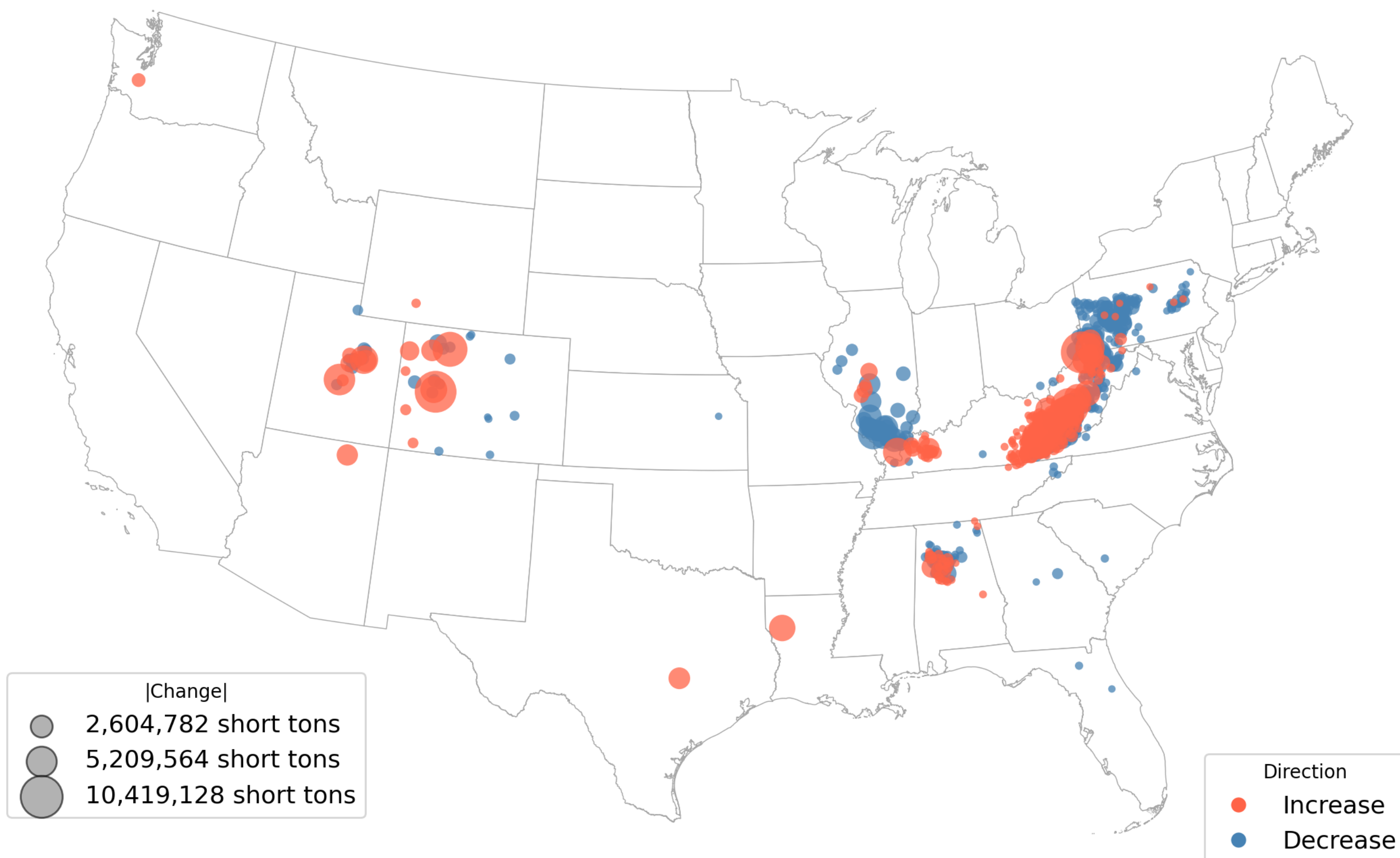

Figure 5: Change in coal production by watershed, 1985 to 2005

Notes: Circles are centered on watersheds with at least one active coal mine between 1985 and 2005 and are sized proportionate to the change in total coal production over the period. Red circles indicate an increase in production and blue circles indicate a decrease.

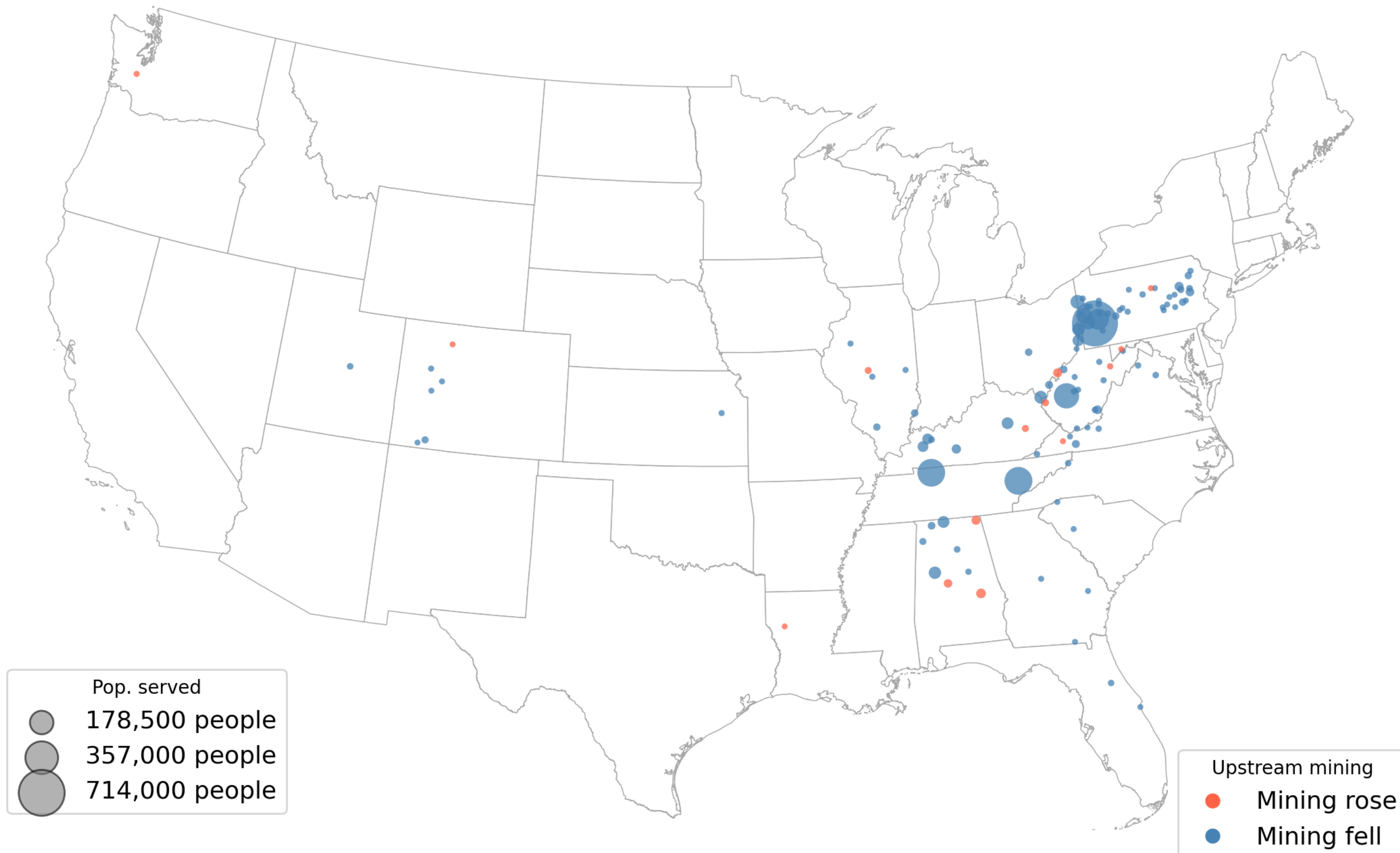

Figure 6: Change in upstream coal production weighted by downstream population

Notes: Circles are centered on coal-mining watersheds located directly upstream of a water utility intake and are sized proportionate to the population served by the downstream utility. Red circles indicate an increase in production and blue circles indicate a decrease. Watersheds with no water utility directly downstream are omitted. The total population one watershed downstream of an active coal mine was 1,257,836 in 1985 and 218,874 in 2005, a net decrease of 1,038,962.

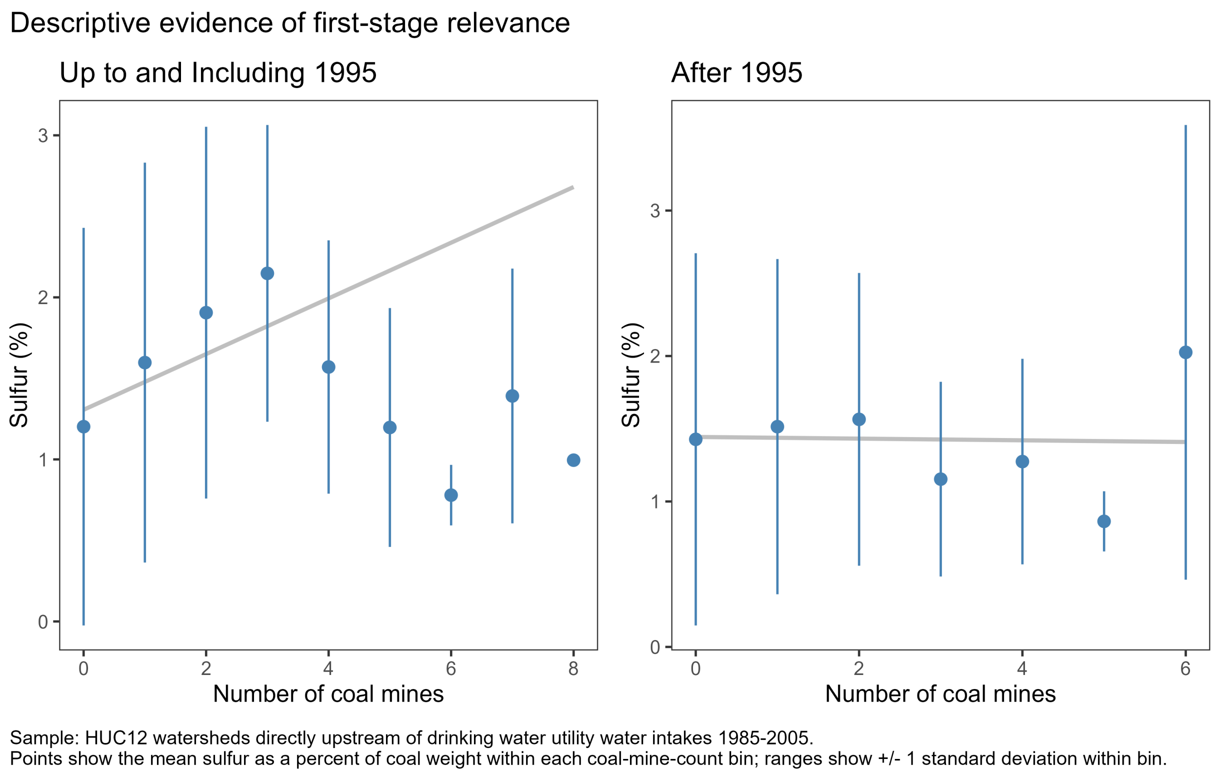

Figure 7: Coal mines and coal sulfur content, before and after 1995

Notes: Each point is a watershed-year observation. Lines are separate linear fits of the number of coal mines on coal sulfur content for the periods before and after 1995.

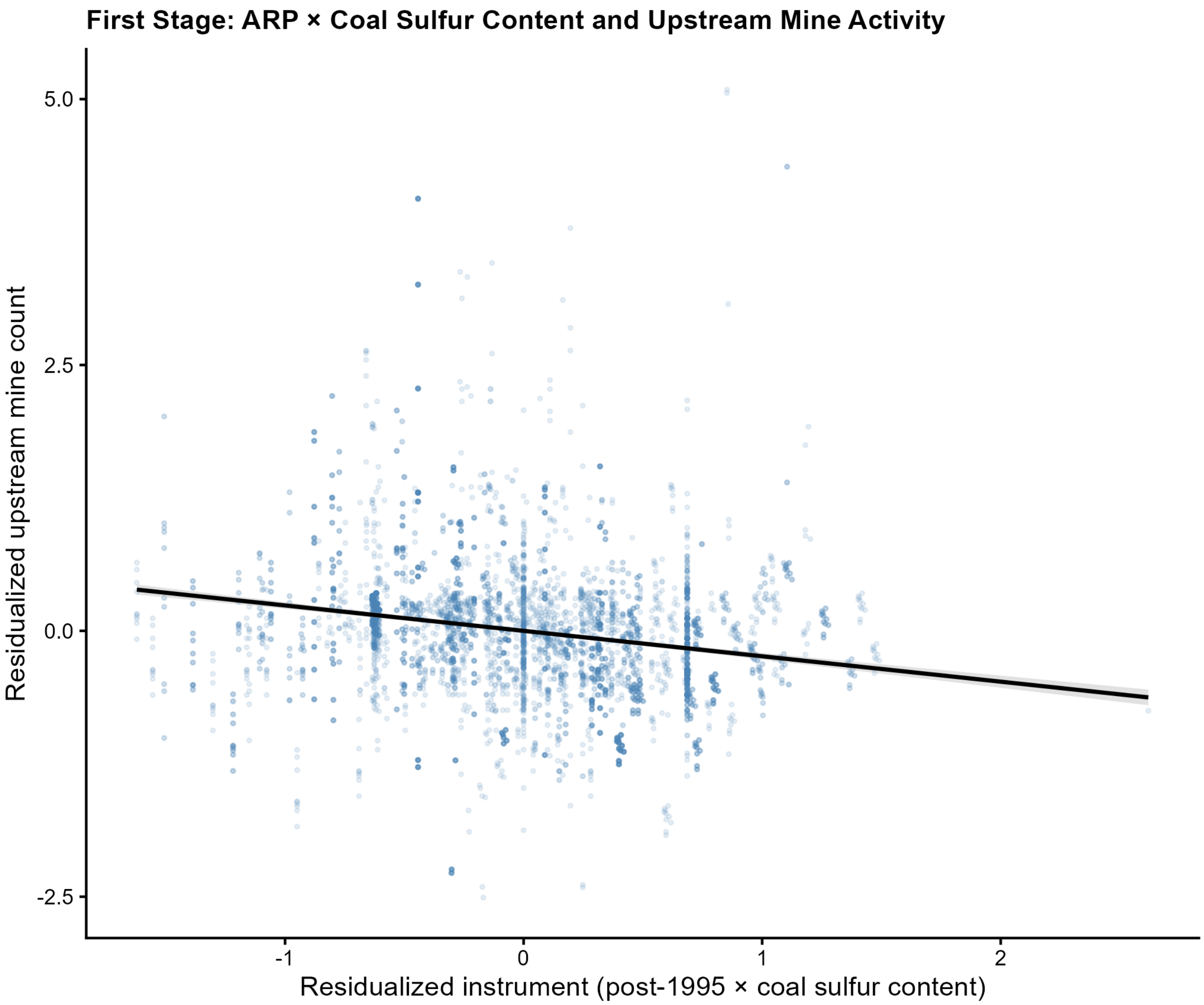

Figure 8: Residualized first-stage relationship

Notes: Each point is a utility by year observation for the sample strictly downstream of a coal mine, 1985 to 2005. Both axes show residuals from separate regressions of each variable on utility and year fixed effects, retaining only within-utility variation over time. The vertical axis residualizes the number of upstream coal mines; the horizontal axis residualizes the instrument, the post-1995 indicator interacted with the sulfur content of upstream coal. By the Frisch-Waugh-Lovell theorem, the slope of the fitted line equals the first-stage coefficient from the two-stage least squares specification.